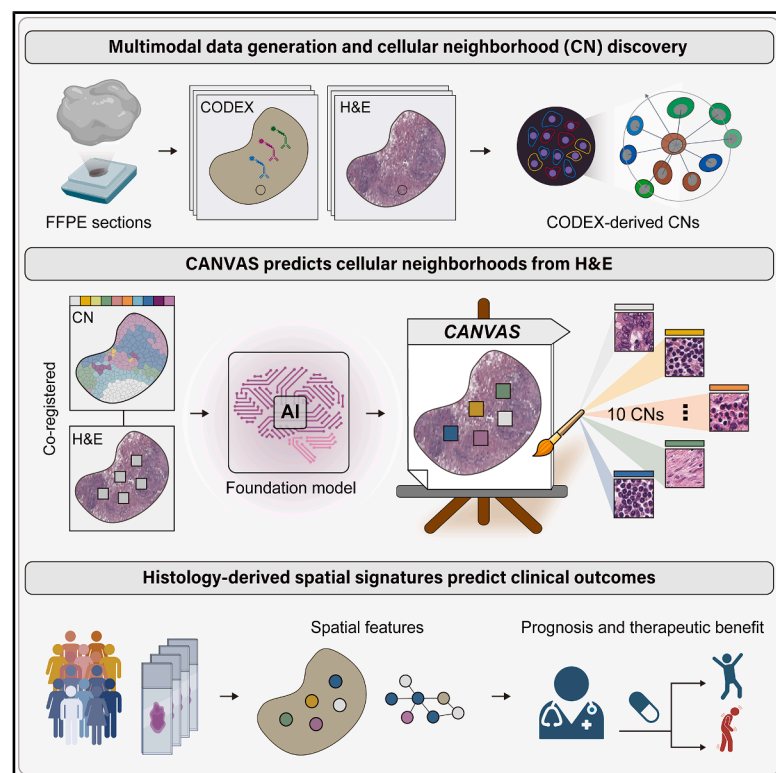


Cellular architecture and neighborhood-informed virtual spatial tumor profiling from histopathology

Graphical abstract



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In brief

CANVAS is a computational platform that decodes spatial tumor ecosystems from routine H&E histopathology to enable population-level prognostic modeling and immunotherapy response.

Highlights

- Single-cell spatial proteomics atlas identifies cellular neighborhoods
- CANVAS predicts CNs from standard histology
- CANVAS enables virtual spatial profiling at the population level
- CANVAS identifies H&E-based spatial signature of immunotherapeutic outcome

Article

Cellular architecture and neighborhood-informed virtual spatial tumor profiling from histopathology

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SUMMARY

The tumor microenvironment (TME) critically shapes disease progression and therapeutic resistance. However, a comprehensive understanding of its spatial architecture remains elusive, and clinical translation is challenging. Here, we present cellular architecture and neighborhood-informed virtual AI-driven spatial profiling (CANVAS), an artificial intelligence platform that infers tumor ecological habitats from hematoxylin and eosin (H&E) histopathology. Built on an atlas of over 18 million cells profiled by 41-plex spatial proteomics across 457 patients with non-small cell lung cancer, CANVAS establishes 10 reproducible cellular neighborhoods (CNs) capturing conserved spatial organization of the TME. Through multimodal alignment and foundation-model-based morphological encoding, CANVAS predicts CN-anchored habitat structures from H&E slides and enables clinical evaluation in over 5,000 patients spanning 9 cancer types. Across patient cohorts, CANVAS supports prognostic modeling, spatial ecotype stratification, and immunotherapy outcome prediction. These results establish CANVAS as a clinically scalable platform for spatial profiling, bridging single-cell analysis to population-level insight and enabling precision oncology.

INTRODUCTION

The tumor microenvironment (TME) is a major determinant of cancer progression and therapeutic response, particularly in non-small cell lung cancer (NSCLC), a clinically heterogeneous malignancy.^{1–3} The TME is a complex ecosystem comprising multiple cell types and distinct cellular states operating through dynamic interactions across broad scales.⁴ Within this landscape, various cell types spatially organize into ecological habitats that represent local niches defined by coordinated phenotypes and functional programs.⁵ However, how multicellular architecture organizes into prognostically and therapeutically relevant spatial habitats remains poorly defined. Addressing this challenge requires technology platforms that can spatially resolve cellular distributions and tissue-level organizations across patient cohorts.^{6,7}

Spatial proteomic technologies have enabled spatially resolved, high-dimensional mapping of proteins in intact tissue.^{8–10} Imaging-based platforms such as co-detection by indexing (CODEX) can resolve cell states, spatial hierarchies, and functional programs at single-cell resolution.^{9,11} Using CODEX, previous studies have uncovered spatial relationships among tumor, immune, and stromal cells, which are inaccessible with traditional bulk profiling.^{11–13} In NSCLC, spatial features including immunosuppressive stromal niches and organized tumor-immune interfaces have been shown to correlate with prognosis^{14,15} and immunotherapy response.¹⁶

Despite its analytical power, the clinical deployment of spatial proteomics is hindered by high cost, technical complexity, and limited scalability. Moreover, most studies on spatial proteomics are confined to a small number of samples, which impact the robustness and generalizability of the findings. By contrast,

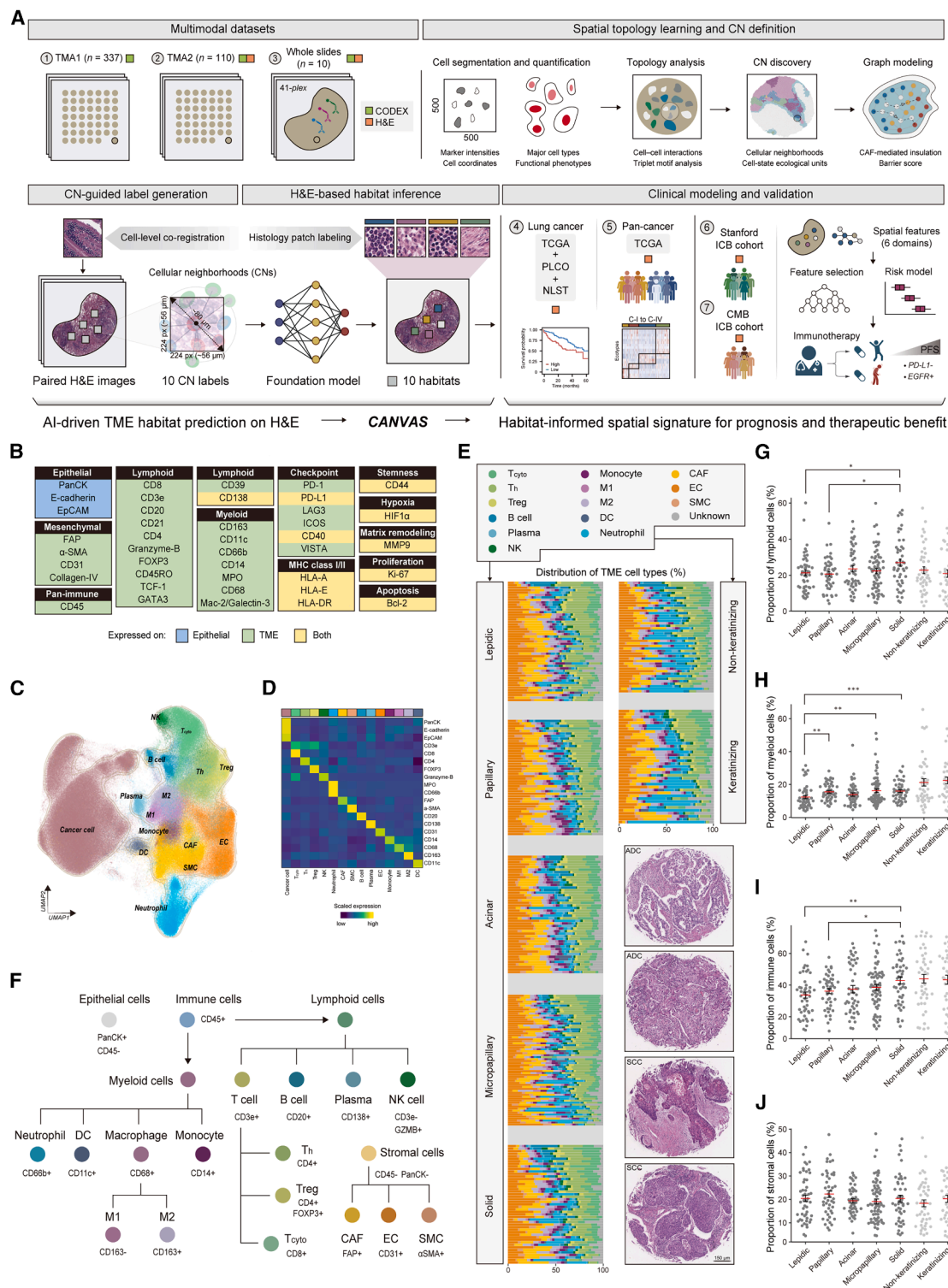


Figure 1. Immune and stromal cell composition across lung cancer histologies

(A) Study design and workflow.

(B) The 41-marker CODEX panel spanning epithelial, lymphoid, myeloid, stromal, and functional programs.

(C) Uniform manifold approximation and projection (UMAP) visualization of single-cell CODEX profiles, with 15 major cell types assigned by lineage-guided annotation. Abbreviations: Tc, cytotoxic T cell; Th, T helper cell; NK, natural killer cell; Treg, regulatory T cell; M1/M2, macrophage subtypes; EC, endothelial cell; SMC, smooth muscle cell; DC, dendritic cell; CAF, cancer-associated fibroblast.

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hematoxylin and eosin (H&E) histopathology remains the clinical standard for cancer diagnosis, offering universal accessibility and compatibility with clinical workflows.^{17,18} While H&E captures gross tissue architecture and cellular morphology,¹⁹ it does not measure protein expression and cannot directly resolve detailed molecular circuits. This trade-off presents a major barrier to translating spatial biology into precision oncology.⁷

Bridging this gap requires a framework that integrates the molecular depth of spatial proteomics with the accessibility of histopathology. A central challenge lies in utilizing deep learning to identify biologically grounded and clinically informative spatial domains from whole-slide histopathology images. Although recent studies have shown the feasibility of inferring spatial protein expression from H&E images, the prediction accuracy is moderately low.^{20–23} More importantly, these models lack the granularity needed to recover functionally coherent spatial structures.

To address this problem, we first delineated the single-cell spatial landscape of the lung TME by profiling over 18 million cells via 41-plex CODEX imaging across 457 tumors. We then constructed a taxonomy composed of 10 cellular neighborhoods (CNs), which capture the spatial topologies of tumor, immune, and stromal architectures conserved across histologic subtypes. Building on this taxonomy, we developed cellular architecture and neighborhood-informed virtual AI-driven spatial profiling (CANVAS), an AI platform that predicts CODEX-defined CNs based on standard H&E histopathology. CANVAS leverages a pretrained pathology foundation model for robust feature extraction, followed by multiscale spatial analysis to uncover informative ecological features. Applied to H&E slides from over 5,000 patients across 9 cancer types, CANVAS enables prognostic modeling, molecular subtype stratification, and immunotherapy outcome prediction. Together, these findings establish CANVAS as a clinically scalable platform that bridges spatial proteomics and histopathology, with a potential for enabling the identification and translation of novel spatial biomarkers for precision oncology.

RESULTS

Distinct immune and stromal remodeling in lung cancer histologic subtypes

To spatially resolve the cellular architecture of the lung TME, we performed high-dimensional CODEX imaging on tumor samples from 457 NSCLC patients, comprising 2 tissue microarray (TMA) cohorts and 10 whole-slide sections with matched H&E staining of the same tissue section (Figure 1A). We used a validated 41-plex antibody panel targeting cell lineage and functional markers, enabling the classification of four major cellular compartments including epithelial, lymphoid, myeloid, and stromal,

with further resolution of intra-lineage phenotypic diversity (Figure 1B; Table S1). The panel included markers representing key functional programs such as proliferation, stemness, hypoxia, antigen presentation, and immune checkpoint activity. This enabled fine-grained annotation of cellular states, which is essential for decoding the spatial and functional heterogeneity of the TME.

We profiled and annotated tumor samples from 337 patients in TMA cohort 1, using a canonical marker-guided, unsupervised lineage classification framework (see STAR Methods). This process allowed us to resolve 14 distinct microenvironmental cell populations in addition to epithelial tumor cells (Figures 1C–1F and S1A). Overall, endothelial cells (ECs) constituted the most abundant non-epithelial population, accounting for 10.4% of all cells. This was followed by T helper cells (Th, 9.6%), cancer-associated fibroblasts (CAFs, 8.2%), and neutrophils (6.5%) (Figure S1B).

Comparative analysis of the two major histologic subtypes of NSCLC revealed distinct differences in microenvironmental composition between lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC). Immune cell infiltration was moderately higher in LUSC (43.7%) compared with LUAD (37.8%; $p < 0.01$), primarily due to an increased proportion of myeloid cells (21.8% vs. 14.8%; $p < 0.001$), while lymphoid and stromal compartments were largely similar (Figure S1C). At the individual cell level, a more detailed analysis revealed that neutrophils and M2 macrophages (M2) were more abundant in LUSC, whereas LUAD exhibited relative enrichment of Th cells, plasma cells, smooth muscle cells, and dendritic cells (DCs) (Figures S1B and S2A).

More granular histologic subtyping further reflected distinct patterns of cellular composition within the TME. In LUAD, progression from lepidic to solid patterns was associated with an increase in lymphoid lineages, while LUSC showed minimal variation in the relative abundance of major TME compartments across subtypes (Figures 1G–1J). Single-cell analysis further delineated subtype-specific trends. In LUSC, keratinization was associated with a significant expansion of CAFs (9.3% vs. 7.1%, $p < 0.05$) accompanied by a decreased infiltration of B cells (2.1% vs. 3.7%, $p < 0.05$) (Figure S2B). Conversely, LUAD demonstrated an increase in adaptive immune constituents with increasing histologic grade, including cytotoxic T (Tc) cells (8.0% vs. 4.5% in solid vs. lepidic, $p < 0.01$), B cells (3.4% vs. 2.3%, $p < 0.01$), and regulatory T cells (Tregs) (2.4% vs. 1.5%, $p < 0.01$) (Figure S2B).

To investigate the association between TME composition and clinicopathologic features, we quantified cell-type proportions in each sample and integrated them with patient-level metadata, including demographics, tumor characteristics, molecular alterations, histologic subtypes, and survival outcomes (Figures 2A

(D) Heatmap of scaled expression of lineage markers across the 15 annotated cell types.

(E) Patient-level distribution of 14 non-epithelial cell types across ADC and SCC, shown as a waterfall plot. Representative H&E images illustrate histologic architecture.

(F) Schematic of hierarchical lineage annotation into epithelial, immune, and stromal compartments, using canonical and marker-defined features.

(G–J) Relative patient-level proportions of lymphoid (G), myeloid (H), total immune (I), and stromal (J) compartments in ADC and SCC.

Data shown as mean $\pm$ s.e.m. Mann-Whitney U test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

See also Figure S1.

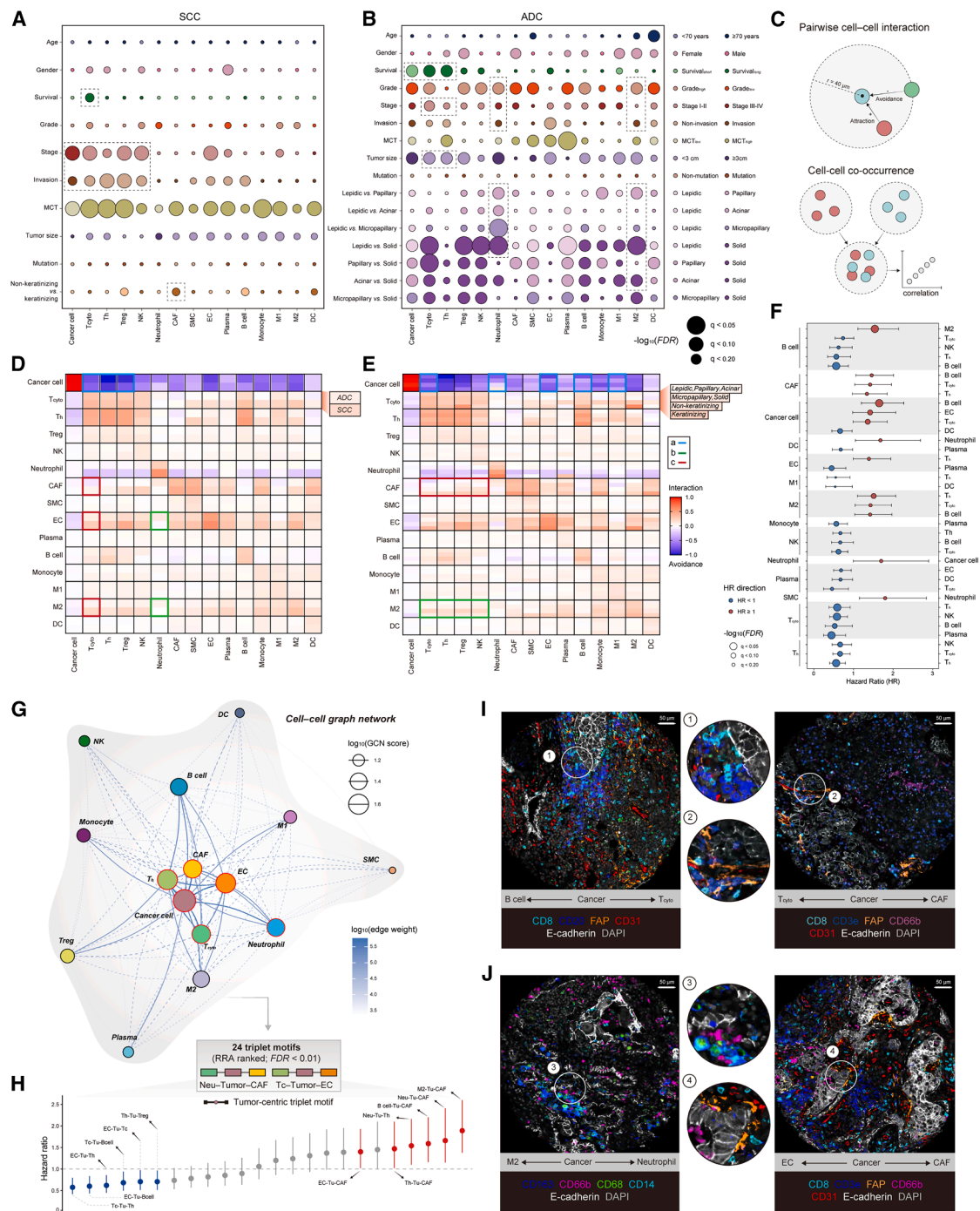


Figure 2. Spatial cell-cell interaction networks and topological rewiring in the NSCLC TME

(A and B) Associations between cell-type proportions and clinicopathologic variables in SCC (A) and ADC (B). Bubble plots show significance ($-\log_{10}FDR$) and direction of association. Mutation status includes epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase (ALK), ROS proto-oncogene 1 (ROS1), or KRAS proto-oncogene (KRAS) alterations. MCT, mast cell tryptase percentage.

(C) Schematic of spatial interaction and co-occurrence analyses.

(D and E) Permutation-derived spatial interaction matrices showing scaled pairwise interaction landscapes across major NSCLC subtypes (ADC and SCC; D) and histologic subtypes (E), with ADC grouped into low-grade and high-grade tumors and SCC stratified into keratinizing and non-keratinizing subtypes.

(F) Forest plot of associations between cell-cell interaction strength and OS using Cox proportional hazards models.

(G) GNN-derived cell-cell topological network constructed from spatial cell graphs across the NSCLC cohort.

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and 2B). In LUSC, cancer cells were more abundant in advanced-stage tumors and those exhibiting lymphovascular or pleural invasion, whereas Tc, Th, and natural killer (NK) cells were enriched in early-stage tumors without pathologic evidence of invasion, reflecting a cytotoxic immune-infiltrated TME. CAFs exhibited selective expansion in keratinizing squamous cell carcinomas (SCCs), suggesting stromal-driven modulation of squamous epithelial differentiation²⁴ (Figure 2A). In LUAD, Tc and Th cells were enriched in solid-predominant histology, early-stage disease, and smaller tumors, where their presence pointed toward an active adaptive immune landscape and correlated with a favorable prognosis (Figure 2B). In poorly differentiated tumors, neutrophils and M2 were enriched in subgroups with higher invasive potential, consistent with pro-tumorigenic inflammation and immunosuppressive remodeling associated with aggressive disease phenotypes²⁵ (Figure 2B). Although TME compositional features correlated with clinical parameters, their limited prognostic utility suggested that additional higher-order spatial relationships may contribute to tumor progression.²⁶

Spatial topology sculpts immunoregulatory architecture in lung cancer

We delineated the intercellular spatial interaction network of the lung TME with a focus on subtype-specific differences in cellular organization. To do this, we applied a permutation-based approach to quantify pairwise spatial interactions between cell types, complemented by co-occurrence analysis at the sample level (Figure 2C; see STAR Methods). In LUAD, tumor cells were spatially insulated from Tc and Th cells, consistent with limited cytotoxic engagement, whereas neutrophils were preferentially localized near M2 and ECs, forming interaction patterns associated with immunosuppressive and angiogenic remodeling (Figure 2D, boxes a and b). In LUSC, Tc and Th cells were more frequently positioned adjacent to stromal and myeloid compartments, including CAFs, M2, and ECs (Figures 2D, box c and S2C), suggesting a subtype-specific redistribution of immune effectors away from tumor cells.

We further evaluated how intercellular networks evolve with histologic progression. In LUAD, compared with other histologic patterns, high-grade patterns (micropapillary and solid) showed increased crosstalk between cancer cells and neutrophils, ECs, M1 macrophages, and B cells in spatial interaction analysis. This was accompanied by diminished tumor-Tc cell interactions despite increased Tc cells in high-grade tumors (Figure 2E, box a). The spatial proximity of neutrophils, ECs, and tumor cells suggests a pro-metastatic vascular gateway that may facilitate vascular invasion and tumor dissemination.²⁷ In high-grade LUSC (non-keratinizing), spatial interactions between M2 and both T lymphocytes and NK cells were enhanced, consistent with the formation of immunosuppressive niches that may constrain effective anti-tumor immunity (Figure 2E, box b). Additionally, co-occurrence analysis showed that DCs tended

to co-aggregate with CAFs, M2, and ECs (Figure S2D, box a). In contrast, keratinizing LUSC exhibited a fibrotic exclusion-like architecture, with CAFs remaining spatially adjacent to Tc, Th, B, and NK cells (Figure 2E, box c). ECs showed strong co-occurrence with M2, CAFs, and B-lineage cells, likely forming perivascular exclusion zones with decreased T cell infiltration (Figure S2D, box b). Despite architectural differences, LUAD and LUSC shared a conserved immunoregulatory TME. Within this shared tumor-immune landscape, subtype-adjusted survival analyses revealed that tumor cell interactions with neutrophils, as well as lymphoid interactions with M2 and CAFs, were associated with adverse prognosis, whereas Tc cell engagement with Th, B, and plasma cells correlated with favorable outcomes (Figure 2F).

To investigate higher-order cellular interactions beyond pairwise proximity and decode spatial topologies, we applied graph neural networks (GNNs) to spatially resolved cell-cell graphs across the NSCLC cohort (see STAR Methods). This analysis revealed a conserved immunoregulatory hub comprising tumor cells, Tc cells, Th cells, M2, CAFs, ECs, and neutrophils (Figure 2G), whose constituent cell types occupied central network positions and orchestrated TME organization. To further dissect the architecture of these regulatory cores, we enumerated all three-cell motifs by integrating GNN-derived node centrality with edge density and ranked them using robust rank aggregation (RRA; see STAR Methods). We identified 24 topologically enriched triplets (false discovery rate [FDR] < 0.01) of which 12 were significantly associated with patient prognosis (Figure 2H). High-risk motifs were dominated by tumor-associated myeloid and stromal triplets (e.g., M2-tumor-CAF, neutrophil-tumor-CAF), reflecting a fibroinflammatory axis of immune suppression, whereas protective motifs incorporated effector lymphocytes and ECs (e.g., Tc-tumor-Th, EC-tumor-B), supporting cytotoxic engagement and vascular accessibility (Figures 2H–2J).

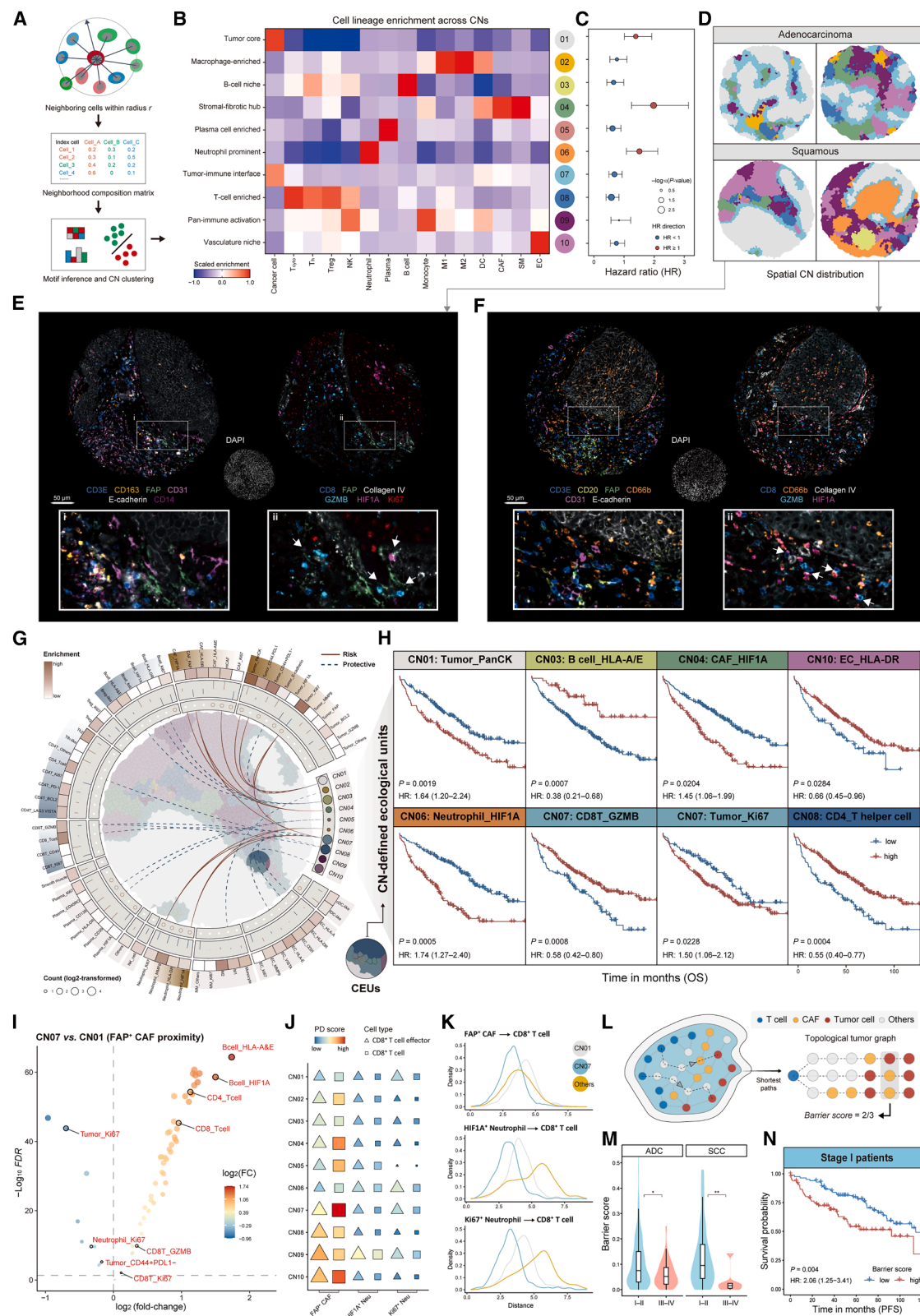
Together, these results underscore the importance of single-cell spatial analysis to capture fundamental TME biology. Our findings suggest that NSCLC progression emerges along two interrelated axes: one driven by shifts in cellular composition and the other sustained by intercellular interactions that establish an immunoregulatory circuitry. These spatial topological architectures were conserved across histologic subtypes, suggesting a unifying structural framework that integrates tumor progression with spatially coordinated immune modulation.

CNs underpin prognostic ecological units in lung cancer

In NSCLC, spatially organized cell states form recurrent multicellular patterns that collectively define hierarchical niches composed of distinct immune and stromal lineages.¹ Owing to their topological recurrence, these multicellular configurations provide a spatially conserved scaffold for decoding TME heterogeneity across tumors.²⁸ To systematically delineate these multicellular networks, we applied an unsupervised clustering algorithm that defined spatially resolved CNs by aggregating

(H) Representative tumor-centered triplet motifs identified by integrating GNN-derived topology and survival associations.

(I and J) Representative CODEX images of tumor-centered triplet motifs, including lymphoid-enriched configurations (I) and myeloid-stromal arrangements (J). Statistical significance was assessed using the Mann-Whitney U test or Cox proportional hazards regression. FDR-adjusted *p* values are shown where indicated. See also Figure S2.



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lineage-annotated cells within a fixed radius (Figures 3A and S3A; see STAR Methods). This analysis revealed 10 CNs that capture archetypal tissue ecologies with distinct tumor-immune or stromal imprints: tumor core (CN01), macrophage-enriched niche (CN02), B cell niche (CN03), stromal-fibrotic domain (CN04), plasma cell cluster (CN05), neutrophil-rich zone (CN06), tumor-immune interface (CN07), T cell compartment (CN08), pan-immune activation zone (CN09), and vasculature niche (CN10) (Figures 3B and 3D–3F).

Our analysis revealed that CN06 (neutrophil-rich zone) was enriched in poorly differentiated, large LUAD tumors, whereas CN08 (T cell compartment) was associated with less invasive, smaller tumors in both subtypes (Figures S3B and S3C). We performed univariate Cox regression analysis to evaluate the prognostic associations of CN-level signals and found that tumor-associated, fibrotic, and neutrophilic structures (CN01, CN04, and CN06) were linked to unfavorable prognosis, whereas B cell, plasma cell, T cell, and tumor-immune interface neighborhoods (CN03, CN05, CN07, and CN08) were linked to favorable clinical outcomes (Figure 3C). These associations were reproduced in an independent NSCLC cohort, supporting the cross-cohort consistency of CN-based distribution patterns (Figures S3D–S3F). Collectively, these findings support that CNs are biologically defined, clinically informative spatial domains that are conserved across histologic subtypes in NSCLC.

To further resolve intra-neighborhood complexity, we defined cell-state-resolved ecological units (CEUs) by expanding the cellular hierarchy to include detailed phenotypic cell states (Figure S3G; Table S3). Quantification of these refined phenotypes within each CN revealed how distinct functional programs shape the cellular composition and organization of spatial niches. Correlation of CEU profiles with clinical trajectories uncovered robust subtype-associated patterns and prognostic associations (Figure 3G). A dichotomous pattern was evident, where CEUs defined by pro-tumorigenic and fibroinflammatory programs within CN01, CN04, and CN06 were correlated with diminished survival expectancy. Conversely, lymphoid-endothelial CEUs located in CN03, CN08, and CN10 were preferentially associated with improved survival (Figures 3G and 3H).

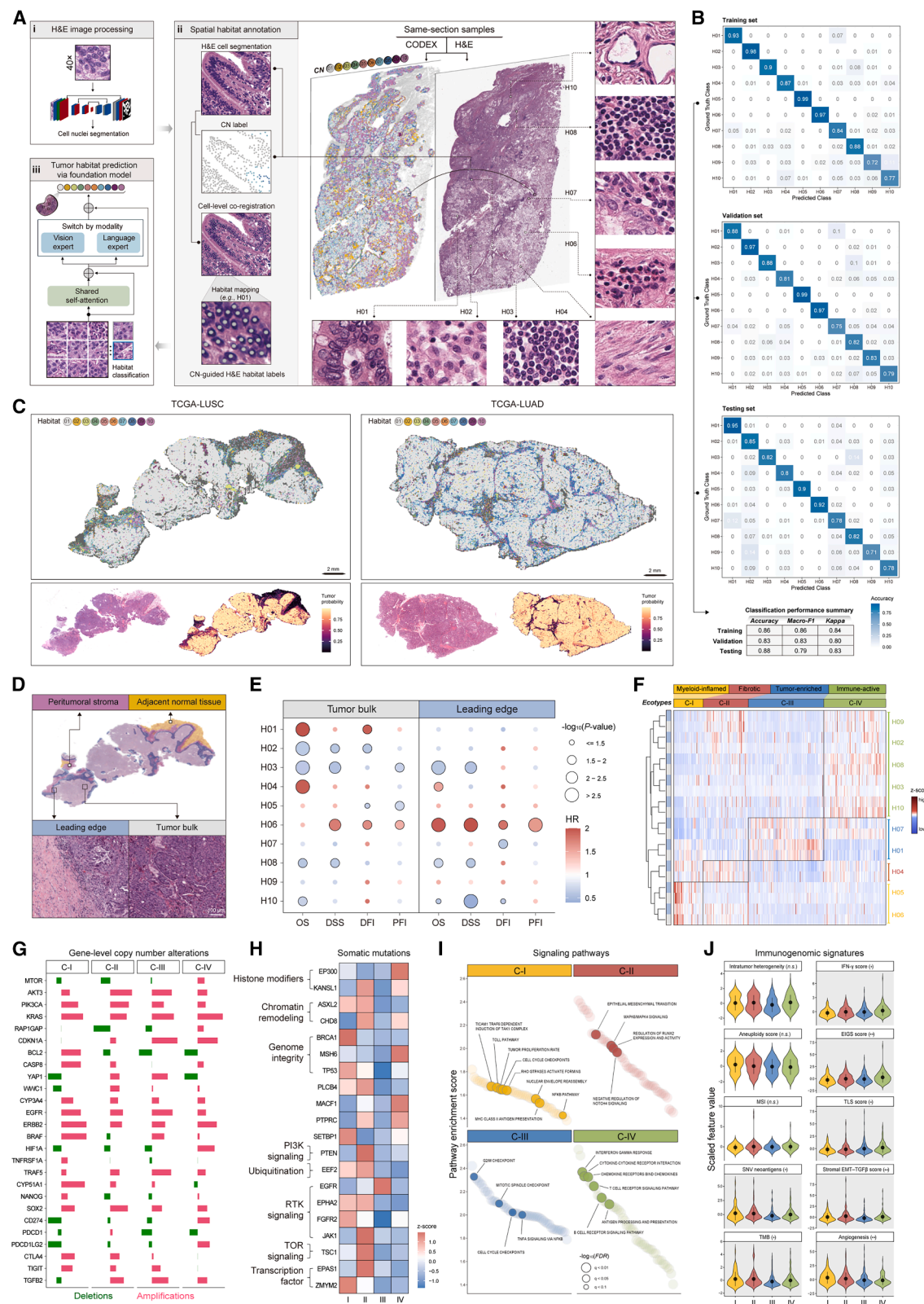
At the cell-state level within CEUs, CN01 was enriched in PanCK⁺ tumor cells, CD44⁺ stem-like subsets, and Ki67⁺ proliferative populations, whereas E-cadherin⁺ tumor cells marked a more preserved epithelial state with limited epithelial-mesenchymal transition (EMT).^{29–31} Within CN04, the accumulation of matrix-associated CAFs (mCAFs) and hypoxia-inducible factor 1- α (HIF1A)⁺ fibroblasts reflected a desmoplastic niche with a hypoxia-driven fibrotic program (Figure 3G). CN06 was dominated by activated neutrophil states, including Ki67⁺/HIF1A⁺ subsets, consistent with a hypoxic and potentially pro-tumorigenic neutrophil program associated with worse outcomes (Figures 3G and 3H).³² In contrast, immunocompetent ecological units demonstrated features of coordinated immune surveillance. CN03 featured an enrichment of B cells expressing human leukocyte antigen (HLA)-A/E and HLA-DR, supporting antigen presentation and humoral immunity within the tumor milieu.³³ CN08 was enriched with cytotoxic CD8⁺ T lymphocytes expressing granzyme B and CD4⁺ Th cells, consistent with sustained effector activity and adaptive immune activation³⁴ (Figures 3G and 3H).

Analysis of CN07 revealed a spatially distinct CEU at the tumor-immune interface, characterized by the coexistence of pro- and anti-tumor features (Figures 3E–3G). Compared with tumor core CN01, this niche uniquely integrated CD8⁺ T cells and their granzyme B (GZMB)⁺ effector subset with Ki67⁺ tumor cells, HIF1A⁺/Ki67⁺ neutrophils, and fibroblast activation protein (FAP)⁺ CAFs, establishing a microenvironment marked by ambivalent immunomodulation (Figures 3G, 3I, S4A, and S4B). To investigate the structural logic of this composite microenvironment, we resegmented all tissue sections using CN boundaries and quantified a spatial metric termed proximity density between CD8⁺ Tc cells and key cellular subsets, including FAP⁺ CAFs and HIF1A⁺/Ki67⁺ neutrophils (see STAR Methods). Proximity density reflects the degree of spatial co-localization normalized by the local abundance of the interacting cells within each CN. Among all CNs, CN07 exhibited the highest proximity density between CAFs and Tc cells (Figure 3J), suggesting a concentrated spatial association within this niche and indicating a dense, spatially restricted interface. These immune-stromal

Figure 3. CNs define prognostic ecological units and spatial immune barriers in NSCLC

- Schematic of CN definition from local cell-type composition within a fixed radius.
 - Heatmap showing scaled lineage enrichment across the 10 CNs, delineating spatial compartments defined by major cell lineages.
 - Forest plot of hazard ratios from univariable Cox analysis of individual CN for OS.
 - Representative CN Voronoi maps from four samples showing CN distributions across ADC and SCC.
 - Representative CODEX images illustrating CN spatial organization based on marker co-expression patterns.
 - Circos plot summarizing the distribution and prognostic significance of cell phenotypes across CN-defined CEUs. Five concentric tracks are shown from the outermost to the innermost: (1) phenotype identity; (2) CEU relative enrichment; (3) scaled log₂ fold changes between ADC and SCC, with blue bars indicating SCC enrichment; (4) statistical significance of subtype differences, where red points denote FDR < 0.01; and (5) representative CEUs with prognostic associations within each CN. An adjacent bubble plot summarizes the number of prognostically significant CEUs (log₂-transformed) per CN.
 - Kaplan-Meier curves for OS of eight selected CEUs.
 - Volcano plot comparing differential proximity of FAP⁺ CAFs to other cell types between CN07 and CN01.
 - Scaled spatial proximity density of FAP⁺ CAFs or HIF1A⁺/Ki67⁺ neutrophils with CD8⁺ T cell subsets across CN domains.
 - Density plots of intercellular distances between FAP⁺ CAFs or HIF1A⁺/Ki67⁺ neutrophils and CD8⁺ T cell across CN01, CN07, and other CNs.
 - Schematics of barrier score calculation based on shortest spatial paths linking Tc cells and tumor cells under CAF-defined spatial constraints.
 - Barrier score distributions in LUAD and LUSC stratified by early- and advanced-stage disease.
 - Kaplan-Meier curves of PFS comparing high and low barrier score groups in stage I NSCLC.
- Boxplots show median, IQR, and 1.5 $\times$ IQR whiskers. Statistical significance was assessed using log-rank tests, Cox regression, or Mann-Whitney U tests. **p* < 0.05, ***p* < 0.01.

See also Figures S3 and S4.



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pairs were positioned at significantly shorter intercellular distances in CN07 than in tumor core CN01, suggesting a compact stromal architecture associated with local Tc cell exclusion at the invasive margin (Figures 3K and S4C).

To quantify the extent of stromal insulation, we computed a barrier score adapted from topological tumor graphs, which measures the degree to which stromal elements, particularly CAFs, spatially interpose between CD8⁺ T cells and proliferating tumor cells (Figure 3L). The triplet-derived barrier score showed a stage-associated pattern, with relatively higher values in early-stage tumors (stages I and II) than in advanced-stage tumors (stages III and IV) (Figures 3M and S4D). This pattern suggests that CAF-mediated insulation is more apparent in early lung cancer, consistent with the presence of a more intact desmoplastic rim in which boundary-localized CAFs and aligned extracellular matrix (ECM) programs can enforce T cell exclusion at the tumor-stroma interface.³⁵ Clinically, elevated barrier scores were associated with worse progression-free survival (PFS) in NSCLC (Figure S4E), particularly among stage I patients ($p = 0.004$; hazard ratio [HR] = 2.06; 95% CI, 1.25–3.41; Figure 3N).

We next expanded the analysis to refined phenotypic cell states to identify more granular CNs. Using this expanded taxonomy, we identified 20 distinct CNs representing discrete cellular niches. Among these, CN04 delineated a hypoxic-proliferative domain characterized by the co-localization of LAG3⁺ CD4⁺ Th cells co-expressing VISTA, HIF1A⁺ neutrophils, and Ki67⁺ tumor cells. CN17 defined an immunoregulatory niche enriched for regulatory B cell (Breg)-like cells, Tregs, Th2-like cells, and PD-1⁺ CD4⁺ T cells (Figures S4F and S4G). Both CN04 and CN17 showed adverse prognostic effects, with their CN-level signals correlated with poor patient outcomes (Figures S4I and S4J). Although the 20-CN model resolved finer-grained niches, the original 10-CN model showed greater compositional purity and lineage homogeneity, supporting a more biologically coherent representation of the lung TME (Figure S4H). Overall, these results establish CNs as spatially resolved tumor architectures, wherein cell-state ecological units bridge functional diversity with a multicellular structure to inform clinical outcomes.

CANVAS enables prediction of tumor habitats from whole-slide histopathology

By defining CNs as spatially recurrent ecological domains, we established a taxonomy that captures TME heterogeneity through conserved cell-state programs and architectural pat-

terns. While this provides important insight into TME biology, CODEX-based spatial proteomics is not practical for routine clinical applications. H&E histopathology, however, is widely used in clinical diagnosis, and advanced computational modeling offers an opportunity to reveal the subtle relationships between molecular phenotypes and morphological patterns.^{36,37} Motivated by this, we developed CANVAS, which predicts tumor ecological habitats from standard H&E histology and comprises two integral modules (Figure 4A; see STAR Methods). First, we adapted a vision-language foundation model pretrained on millions of paired histopathology images and semantic annotations to predict CODEX-defined CNs as spatial habitats directly from H&E slides. Second, we constructed clinical outcome prediction models based on habitat-centric spatial features to capture biologically relevant patterns and enable histopathology-based patient stratification. Together, these modules support scalable and clinically translatable spatial diagnostics for precision oncology.

The first module of the CANVAS approach comprised four core components: (1) annotation of CODEX-defined CNs as the ground truth for H&E-based habitat prediction, (2) automated nuclear segmentation and localization on H&E-stained sections, (3) single-cell spatial co-registration of CODEX and H&E images on the same tissue to align CN identities with histology, and (4) foundation model-driven prediction of CODEX-defined CNs using H&E as input (Figure 4A; see STAR Methods). This pipeline enables the precise transfer of spatially informative CNs from high-plex CODEX to standard H&E images, supporting histology-native spatial diagnostics.

To support model development, we curated a multimodal dataset of CODEX/H&E image patches (224 × 224 pixels) from whole-slide images (WSIs) and TMA, with patch labels assigned through cell-level CODEX-H&E co-registration (see STAR Methods). After quality control, each patch was assigned to one of 10 CN labels based on CODEX analysis. The WSI cohort was split into training ($n = 8$ WSIs) and validation ($n = 2$ WSIs) sets, whereas the TMA cohort 2 was used as an independent test set. As a result, CANVAS demonstrated robust patch-level generalizability across training, validation, and independent test sets, achieving high classification performance with accuracy exceeding 0.83, F1 scores above 0.79, and Kappa coefficients above 0.80 (Figures 4B and S5A). In benchmarking analysis, CANVAS consistently outperformed both established pathology foundation models and traditional convolutional

Figure 4. CANVAS enables accurate prediction of tumor habitats from whole-slide histopathology with prognostic and immunogenomic correlates

- (A) Schematic of CANVAS predicting CODEX-derived CN annotations from paired H&E images.
- (B) Confusion matrices showing CANVAS classification performance in the training, validation, and testing sets.
- (C) Representative CANVAS-derived habitat maps, tumor probability maps, and H&E slides from two TCGA samples.
- (D) Representative H&E regions defining tumor bulk and leading-edge tissue compartments.
- (E) Bubble plots showing survival associations of habitat profiles in tumor bulk and leading-edge regions in the TCGA-LUAD cohort.
- (F) Consensus clustering of habitat compositions identifying four spatial ecotypes (C-I to C-IV) across lung cancer tissues.
- (G) Gene-level copy-number alterations associated with spatial ecotypes, summarized as log2 odds ratios.
- (H) Heatmap of somatic mutation enrichment across spatial ecotypes, with mutation frequencies Z score normalized.
- (I) Representative pathways differentially enriched across the four spatial ecotypes.
- (J) Distribution of immunotherapy-related molecular features across ecotypes.

Statistical significance was assessed using Cox regression or ANOVA. * $p < 0.05$; ** $p < 0.01$; n.s., not significant.

See also Figures S5–S7.

neural networks for habitat prediction ($p < 0.01$; Figure S7A). Beyond these benchmarks, CANVAS maintained strong performance for varying H&E resolutions, with comparable accuracy between $\times 40$ and $\times 20$ magnifications (Figure S7B). Finally, CANVAS remained robust across different patch extraction strides, with non-overlapping tiling yielding performance comparable to tiling with up to 75% overlap and accuracy consistently exceeding 0.85 (Figure S7C; see STAR Methods).

These results established the technical robustness of CANVAS for habitat inference from routine H&E images. We next applied CANVAS to three NSCLC cohorts, including The Cancer Genome Atlas (TCGA), Prostate, Lung, Colorectal and Ovarian Cancer Screening Trial (PLCO), and National Lung Screening Trial (NLST), covering both LUAD and LUSC histologies. CANVAS accurately mapped spatial habitat architectures and identified recurrent habitat-level patterns linked to clinical outcomes (Figures 4C–4E). In TCGA-LUAD, higher levels of H01, H04, and H06 were associated with poor survival, whereas H03, H05, and H08 were associated with favorable prognosis, consistent with our CODEX-based findings (Figure 4E). These associations were refined by regional partitioning into tumor bulk and leading-edge compartments (Figures 4D and 4E; see STAR Methods). Similar prognostic patterns were observed in the PLCO and NLST cohorts, where greater representation of H01, H04, and H06 were generally associated with adverse outcomes, whereas H03 and H08 were associated with favorable prognosis (Figures S6A, S6D, and S6E).

To assess the broader applicability of CANVAS, we applied the model to same-section CODEX-H&E data from 17 specimens spanning 12 tumor types. In this benchmark, CANVAS showed concordance between H&E-predicted and CODEX-derived habitat profiles across diverse tumor contexts (Figure S7D). Orthogonal validation using same-section spatial transcriptomics (ST)-H&E samples further showed agreement between H&E-predicted and ST-derived habitat profiles (Figure S7E). We then extended the analysis to 3,724 H&E slides from 8 TCGA cancer types, revealing CANVAS-inferred habitats showed recurrent but cancer type-dependent prognostic associations beyond lung cancer (Figure S7F).

CANVAS defines spatial ecotypes associated with immunogenomic features

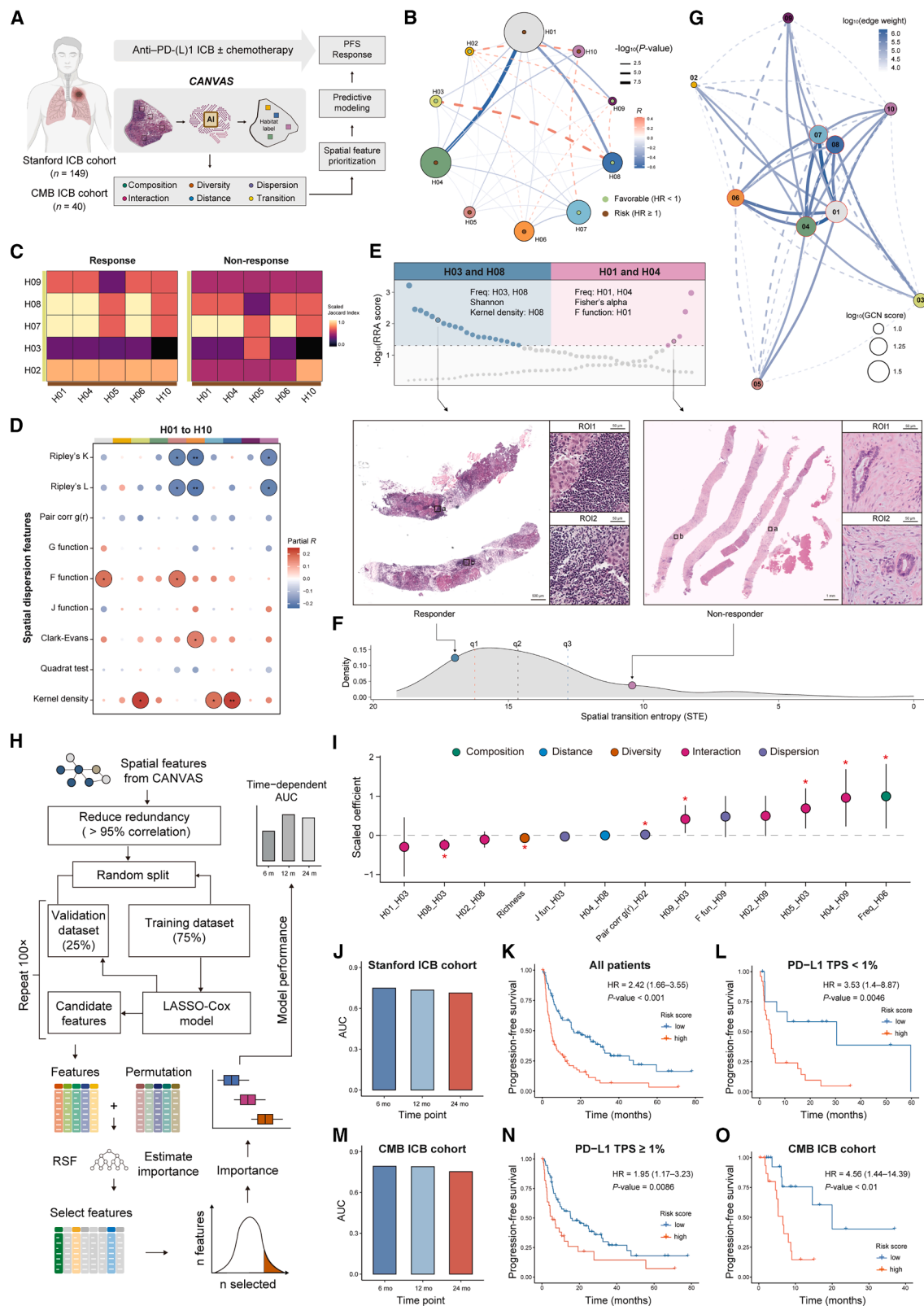
To assess relationships between histology-inferred tumor habitats and immunogenomic features, we compared CANVAS-predicted spatial architectures with transcriptome-deconvoluted cellular composition and signatures in TCGA-LUAD cohort (see STAR Methods). This analysis revealed coordinated enrichment patterns, where H06 correlated with neutrophil abundance, while H03 and H08 clustered with adaptive immune modules enriched for B and T cells (Figures S5B–S5D).

Building on these associations, we performed unsupervised consensus clustering of tumor habitat proportions in the TCGA-LUAD cohort, which revealed four tumor clusters or ecotypes with distinct spatial architecture profiles (Figure 4F). C-I (myeloid-inflamed) was dominated by habitats H05 and H06, indicative of an inflammatory neutrophil-rich and innate-oriented TME. C-II (fibrotic) featured a fibrotic habitat H04 with minimal immune infiltration. C-III (tumor-enriched) was mainly composed

of habitats H01 and H07, characterized by high tumor purity and low stromal and immune content. C-IV (immune-active) exhibited coordinated enrichment of lymphocyte-abundant habitats (H03, H08) (Figure 4F). Although spatial ecotypes showed partial correspondence with established TME classifiers,³⁸ they demonstrated substantial independence from bulk transcriptome-based classifications (Cramér's $V = 0.18$; Figure S5E). These spatial ecotypes stratified survival outcomes across various clinical endpoints (log-rank $p < 0.05$; Figures S6B and S6C) and remained independently prognostic for overall survival (OS) and disease-specific survival (DSS) in multivariable Cox models adjusted for clinicopathologic covariates (Figures S6F and S6G).

To enhance the biological interpretability of CANVAS-defined spatial ecotypes, we integrated each cluster with multi-omic features encompassing somatic alterations, oncogenic signaling, and immunotherapy-relevant transcriptional programs. C-I was characterized by co-amplification of *BCL2* and *BRAF*, consistent with enhanced survival and proliferative signaling.^{39,40} Upregulation of *CYP51A1* further suggested altered cholesterol biosynthesis.⁴¹ C-II exhibited deletions in *MTOR* and *RAP1GAP*, in line with dysregulated phosphatidylinositol 3-kinase (PI3K) and Ras-associated protein 1 (Rap1) signaling, while modest *SOX2* amplification was compatible with increased epithelial plasticity.⁴² C-III, defined by gains in *CDKN1A*, *TRAF5*, and *TGFB2*, reflected a tumor-rich phenotype with active transforming growth factor β (TGF- β) signaling.⁴³ In contrast, C-IV showed amplification of *HIF1A* and immune checkpoint loci (*CD274* and *PDCD1LG2*), denoting a hypoxia-adapted, immune-infiltrated ecotype with potential immunotherapy relevance⁴⁴ (Figure 4G). Complementary analysis of SNV profiles reinforced these ecotypic distinctions. C-I harbored mutations in *BRCA1* and *FGFR2*, consistent with alterations in DNA repair and receptor tyrosine kinase (RTK) signaling.^{45,46} C-II featured alterations in *PTEN*, *JAK1*, *TSC1*, and *EPAS1*, suggestive of disrupted immune- and hypoxia-related signaling.⁴⁷ C-IV was enriched for mutations in *EP300*, *MSH6*, and *MACF1*, associated with immunogenicity, mismatch repair deficiency, and cytoskeletal remodeling.⁴⁸ Conversely, C-III displayed a sparse mutational burden dominated by epidermal growth factor receptor (*EGFR*) mutations, consistent with an EGFR-driven and less immune-infiltrated tumor state⁴⁹ (Figure 4H).

Each CANVAS-defined ecotype exhibited distinct transcriptional programs reflective of its spatially encoded microenvironment. C-I activated Toll-like receptor signaling, consistent with neutrophil-driven innate inflammation and limited lymphoid infiltration.⁵⁰ C-II upregulated mesenchymal reprogramming pathways, including EMT, mitogen-activated protein kinase (MAPK)4/6-associated signaling, and *RUNX2* activity, characteristic of a fibrotic and immunologically quiescent stromal niche.⁵¹ C-III, associated with EGFR mutations and a paucity of lymphoid infiltration, was characterized by heightened expression of cell-cycle regulators, including G2/M checkpoint and mitotic spindle genes, suggestive of a hyperproliferative tumor-rich ecotype.⁵² By contrast, C-IV demonstrated a robust immune activation program encompassing interferon (IFN)- γ response, cytokine-receptor interactions, and lymphocyte receptor pathways,



(legend on next page)

consistent with its densely vascularized and immune-infiltrated microenvironment⁵³ (Figure 4I).

Finally, immunotherapy-associated genomic and transcriptomic signatures further delineated the functional divergence across ecotypes (Table S4). C-IV exhibited enrichment of IFN- γ response (IFN- γ score), expanded immune gene signature (EIGS), and tertiary lymphoid structure (TLS) scores,^{53,54} highlighting a strong adaptive immune response. C-II displayed activation of a stromal EMT-TGF- β axis,⁵⁵ consistent with desmoplastic remodeling and immune exclusion—features linked to resistance to immune checkpoint blockade (ICB). Although both C-I and C-II harbor elevated tumor mutational burdens, their immunosuppressive features likely neutralize neoantigen-driven immunogenicity, thereby impairing the anti-tumor immune response^{56,57} (Figure 4J). These results highlight the therapeutic relevance of the CANVAS-defined spatial ecotypes in the context of immunotherapy.

CANVAS-guided multiscale spatial analysis reveals determinants of immunotherapy benefit

There is an unmet clinical need for reliable predictive biomarkers of immunotherapy response. Traditional biomarkers such as programmed death-ligand 1 (PD-L1) expression lack sufficient predictive power, while genomic and transcriptomic analyses obscure spatial heterogeneity by collapsing into bulk signatures.⁵⁸ CANVAS addresses these limitations through an interpretable AI workflow by extracting habitat-informed multiscale spatial features that reflect ecological diversity, interdomain coupling, and topographic organization, thereby enabling a biologically grounded prediction of therapeutic response (Figure 5A; see STAR Methods).

To translate this spatial modeling paradigm into actionable analytics, we applied CANVAS to a discovery cohort of 149 NSCLC patients treated with immunotherapy (Figure 5A). For each patient, a total of 262 spatially grounded features were extracted (Table S5), which were divided into 6 groups: (1) composition, capturing the relative abundance of each habitat within tumor sections ($n = 10$); (2) diversity,⁵⁹ representing the ecological complexity of habitat mixtures ($n = 6$); (3) spatial dispersion,⁶⁰ capturing habitat-level aggregation and distribution patterns

via multiscale spatial statistics ($n = 90$); (4) interaction, modeling inter-habitat coupling based on pairwise proximity patterns ($n = 100$); (5) distance, quantifying absolute spatial segregation between habitat pairs ($n = 55$); and (6) transition,⁶¹ an entropy-based, non-distance metric reflecting spatial intermixing ($n = 1$) (Table S5; see STAR Methods).

Five habitats (H01, H04, H06, H07, H08) emerged as the most abundant spatial domains in NSCLC. Co-localization analysis revealed recurrent pairing of H03 with H08, consistent with coordinated lymphoid organization, and of H02 with H10, suggestive of immune-vascular interface zones (Figures 5B and S8A). In contrast, H01 was spatially excluded from H04 and H08, reflecting immune-privileged tumor cores insulated from stromal or lymphoid infiltration (Figures 5B and S8A). In the discovery cohort, enrichment of H01 and H06 was associated with inferior PFS, whereas the presence of H02, H07, and H08 delineated immune-active ecosystems predictive of clinical benefit from ICB (Figure S8B). To further resolve habitat-level spatial co-occupancy associated with immunotherapy response, we stratified tumors by ICB outcome and calculated the Jaccard index, which quantifies the degree of shared occupancy between each pair of habitats across tumors. Compared with non-responders, responders showed increased spatial co-occupancy between immune-supportive habitats (H08) and resistance-associated niches (H01, H04, H06) (Figure 5C), suggesting a more permissive spatial interface between immune-active and tumor-associated regions.

We next evaluated the predictive value of habitat-level spatial dispersion. High Ripley's K and L values in H06 and H10 indicated tightly clustered neutrophil-rich and vascular regions, which were linked to shorter PFS, suggesting the formation of inflammatory niches that may impair therapeutic response⁶² (Figures 5D and S8C). Conversely, higher F-function values in H01 and increased kernel density in H03 and H08 correlated with prolonged PFS, associated with favorable spatial organization of tumor and immune-supportive habitats⁶³ (Figure 5D). To identify spatial phenotypes associated with ICB response, we applied RRA to prioritize patients across candidate features, revealing response-linked spatial specificity among prioritized patients (Figure 5E). Among responders, tumors exhibited a

Figure 5. Spatial analysis of H&E-derived tumor habitats identifies a signature predictive of immunotherapy outcomes

- Schematic overview of CANVAS for habitat inference and outcome prediction in immunotherapy cohorts.
 - Integrated habitat co-occurrence and prognostic network in the discovery ICB cohort. Node size denotes habitat abundance, edge width indicates co-enrichment strength, edge color reflects Pearson correlation, and inner node color encodes PFS association.
 - Heatmaps comparing pairwise habitat co-occupancy profiles between responders and non-responders based on Jaccard index analysis.
 - Bubble plots showing associations between habitat-level spatial dispersion features and immunotherapy PFS.
 - RRA-based ranking of samples using spatial features, with representative H&E images showing immune- and stromal-enriched patterns.
 - Density plot of spatial transition entropy across samples in the discovery cohort.
 - GNN-derived habitat connectivity network showing inter-habitat organization.
 - Schematic of feature selection and model construction for PFS prediction.
 - Forest plot of scaled β coefficients for the 13 spatial habitat features retained in the Cox prognostic model.
 - Time-dependent AUCs for PFS prediction at 6, 12, and 24 months in the discovery cohort.
 - (K and L) Kaplan-Meier curves for PFS stratified by the spatial signature in all patients from the discovery cohort (K) and in PD-L1-negative tumors (TPS < 1%; L).
 - Time-dependent AUCs for PFS prediction at 6, 12, and 24 months in the CMB cohort.
 - Kaplan-Meier curves for PFS stratified by the spatial signature in PD-L1-positive tumors (TPS $\geq$ 1%) from the discovery cohort.
 - Kaplan-Meier curves for PFS stratified by the spatial signature in the CMB cohort.
- Statistical significance was assessed using log-rank tests or Cox regression. * $p < 0.05$; ** $p < 0.01$. See also Figure S8.

high abundance of H03 and H08, elevated Shannon diversity, and increased kernel density in H08, consistent with lymphoid-enriched immune organization. In contrast, non-responders were characterized by dominant H01 and H04 habitats together with increased Fisher's alpha, consistent with a heterogeneous but fibrotic and immune-restricted spatial phenotype⁶⁴ (Figure 5E). These two phenotypes also exhibited opposing distributions along the spatial transition entropy ranking, supporting a link between spatial transition complexity and differential ICB responsiveness (Figure 5F). We further assessed the inter-habitat spatial connectivity by conducting GNN-based connectivity analysis, which identified H01, H04, H06, and H08 as central interaction hubs in NSCLC (Figure 5G). Collectively, these multiscale spatial features delineate a complex TME landscape that exerts a substantial impact on immunotherapy benefit.

Habitat-informed spatial signature predicts immunotherapy outcomes

Given the strong association between habitat features and immunotherapy response, we set out to develop a habitat-informed spatial signature that could predict immunotherapy outcomes. In the discovery cohort, we applied CANVAS to pre-treatment H&E images and computed 262 habitat-based spatial features for each patient. We sequentially applied two independent machine learning-based feature selection strategies following redundancy reduction and then trained a multivariable Cox proportional hazards model for PFS (Figure 5H; see STAR Methods).

The final spatial signature included 13 habitat-based features (Figures 5I and S8D). Among these, protective features ($\beta < 0$) primarily reflect immune-activated TME phenotypes. Representative features included co-localization of H08 and H03, suggesting coordinated T and B cell organization and TLS-like structures, together with H01-H03 spatial adjacency and elevated habitat richness, consistent with structured immune ecosystems associated with favorable outcomes^{54,65} (Figure 5I). By contrast, risk features ($\beta > 0$) captured TME architectural mechanisms of resistance to ICBs. High frequency of H06 was associated with the accumulation of neutrophil-rich habitats, consistent with N2-like tumor-associated neutrophil (TAN) features.⁶⁶ H04-H09 interactions indicated fibrotic and immunosuppressive spatial coupling,^{35,67} whereas frequent H05-H03 contact suggested altered humoral immune organization involving B cells and plasma cells.⁶⁸ Close spatial association between H02 and H09, together with elevated pair correlation in H02, was consistent with macrophage-rich immunosuppressive niches and M2-like polarization^{69,70} (Figure 5I).

In the discovery cohort, the spatial signature showed strong performance for predicting PFS after immunotherapy, with time-dependent area under the curves (AUCs) of 0.75, 0.73, and 0.71 for 6-, 12-, and 24-month PFS, respectively (Figure 5J). Notably, the spatial signature outperformed the enrichment of individual spatial habitats, distinguished ICB response groups, and significantly stratified immunotherapy-treated patients by PFS in the discovery cohort (HR = 2.42 [1.66–3.55], $p < 0.001$; Figures 5K, S8E, and S8F). This stratification remained significant across patient subgroups defined by PD-L1 tumor proportion score (TPS; TPS < 1% and TPS $\geq$

1%; Figures 5L and 5N). The spatial signature also identified subgroups with immunotherapy-associated outcomes within EGFR- and KRAS proto-oncogene (KRAS)-mutant tumors, molecular subtypes traditionally considered challenging to ICB^{71,72} (Figures S8G and S8H). The spatial signature remained predictive across therapeutic modalities, including ICB monotherapy and combination ICB-chemotherapy regimens (Figures S8I and S8J). Importantly, the CANVAS-derived spatial signature outperformed established immunotherapy biomarkers (TMB, PD-L1, and TLS), showed improved prediction of PFS when compared with them ($p < 0.05$; Figure S8K), and remained an independent predictor of PFS in multivariable models ($p = 0.002$; Figure S8L).

Finally, we applied CANVAS to an external validation cohort of 40 immunotherapy-treated patients whose pre-treatment H&E images and clinical outcomes were obtained from the Cancer Moonshot Biobank (CMB) Project. The derived spatial signature retained strong predictive performance, with AUCs above 0.75 for 6-, 12-, and 24-month PFS (Figure 5M). It also significantly stratified patients by PFS (HR = 4.56 [1.44–14.39], $p < 0.01$; Figure 5O), underscoring its potential as a clinically relevant, spatially informed biomarker of immunotherapy benefit.

DISCUSSION

In this study, we developed CANVAS, a computational approach that infers tumor ecological habitats from standard H&E histopathology. Through spatial proteomic profiling of over 18 million cells from 457 patients, we identified 10 CNs that capture multicellular architectures in the lung TME. CANVAS infers these CN-anchored spatial structures directly from whole-slide H&E images, enabling histology-native TME profiling. We applied CANVAS to H&E slides from over 5,000 patients and demonstrated its capability for population-scale prognostic modeling, molecular subtype classification, and immunotherapy response prediction. These findings establish AI-driven integration of spatial proteomics and histopathology as a scalable approach for defining spatial ecotypes that inform precision cancer therapy.

We resolved lineage-specific ecological programs in NSCLC, with LUAD progression linked to coordinated B cell and Tc cell infiltration, whereas LUSC, relative to LUAD, was associated with CAF-, M2-, and neutrophil-rich remodeling accompanied by lymphoid depletion, together defining distinct modes of TME reprogramming. Conserved tumor-centric triplet motifs across subtypes further suggested that recurrent higher-order spatial interactions provide a regulatory scaffold for tumor ecological evolution.

To define the interscale logic of spatial architectures, we identified 10 CNs that integrate local cellular programs and topological features into spatially coherent microdomains. Unlike prior studies based on small TMA cores prone to sampling bias, we analyzed both TMAs and whole-slide tissue sections from large patient cohorts, enabling robust CN construction that captures intratumoral heterogeneity while preserving intertumoral diversity. We show that CN06, an immunosuppressive niche enriched in neutrophils, is associated with inferior survival and resistance to anti-PD-1 immunotherapy, supporting TAN-targeted strategies such as CXCR1/2 inhibition with SX-682, which is under

clinical evaluation in NSCLC and colorectal cancer.⁷³ We also identified a multicellular spatial context niche enriched in proliferative tumor cells, Tc cells, and CAFs. Separately, we quantified a CAF-mediated tumor-T cell barrier score associated with worse outcomes in NSCLC. This supports stromal-targeting approaches such as setanaxib,⁷⁴ a NOX1/4 inhibitor being evaluated in combination with pembrolizumab for advanced head and neck squamous cell carcinoma.

At finer resolution, we further identified a prognostically relevant niche in which LAG3⁺VISTA⁺ CD4⁺ Th cells co-localize with HIF1A⁺ neutrophils and Ki67⁺ tumor cells, consistent with a locally exhausted and hypoxia-adapted ecosystem that couples immune suppression with aggressive tumor growth.^{75,76} In NSCLC, LAG-3 restrains T cell activation and promotes immune escape, and therapeutic antibodies targeting this pathway, exemplified by relatlimab, are already being evaluated in combination with PD-1 blockade.⁷⁷ Together, these multiscale spatial niches delineate complementary axes of immunotherapy resistance and provide a rationale for combinatorial treatment strategies that jointly target stromal insulation, neutrophil-mediated immunosuppression, and checkpoint-driven T cell exhaustion.

Although spatial proteomics offers valuable insights into the TME, its clinical translation faces significant challenges including high cost, complexity, and limited scalability. CANVAS captures the relationship between spatially resolved CNs and morphological features derived from H&E through a foundation model pretrained on millions of pathology images.⁷⁸ This is distinct from previous studies that attempted to infer the expression of individual proteins from H&E images. One study applied a generative adversarial network to generate virtual multiplex images of 11 proteins mostly consisting of cell-lineage markers.²² Another recent study used a segmentation-based network to infer the status (high vs. low expression) for 21 proteins.²¹ However, predicting protein expression from H&E is strongly marker dependent and sensitive to staining, acquisition, and assay variability, leading to uneven and often modest performance across markers. By contrast, our study aims to infer tumor ecological habitats—spatial structures with biological interpretation—from H&E at a much higher accuracy exceeding 0.83. This enables robust discovery of spatially informed biomarkers in large-scale patient cohorts.

ICB is a standard treatment for advanced lung cancer, but only a subset of patients derives durable benefit, and robust predictive biomarkers remain lacking. We applied CANVAS to two independent ICB-treated cohorts, and we found that spatial features beyond simple proximity—including intercellular dispersion, diversity, and topological transitions—better distinguished responders from non-responders. Our results show that responsive tumors exhibited B and T cell co-localization and compartmentalized TLS-like immune structures, whereas resistant tumors showed stromal insulation and neutrophil enrichment consistent with previous studies.^{79–81} In addition, we discovered new spatial features, including habitat richness and spatial interaction, which were also associated with ICB response. By integrating these habitat-level patterns, we derived an interpretable spatial signature that outperformed existing biomarkers in predicting ICB outcome.

In conclusion, our study provides a comprehensive characterization of the single-cell spatial landscape of the lung TME, and the computational CANVAS approach provides a biologically informed and clinically scalable platform for histology-native spatial tumor profiling to enhance precision oncology.

Limitations of the study

Our study has several limitations that should be considered when interpreting the results and that point toward future directions. First, CANVAS was trained on paired spatial proteomics and histopathology data in lung cancer, and although validated across multiple cancer types, additional training on more diverse datasets will be necessary to truly generalize the approach in a broader pan-cancer setting. Second, the 10-CN taxonomy is derived from unsupervised clustering of multiplexed imaging data, and while it has been validated through multiple layers, alternative solutions by integrating with additional assays such as spatial transcriptomics could reveal more granular and complementary aspects of TME organization.^{82,83} Third, some of the biological findings such as the association between neutrophil-rich niche and unfavorable outcomes are broadly consistent with established TME patterns. While we view this concordance as supporting the validity of the habitat framework, the more specific mechanistic interpretations (e.g., niche-restricted hypoxia-exhaustion patterns) remain correlative. Functional validation of these spatial ecological hypotheses in experimental models will be an important next step. Finally, the immunotherapy response prediction model was developed in a discovery cohort of 149 patients, and external validation was performed in a modest cohort. These results should be considered hypothesis-generating, and prospective validation in larger cohorts is required before clinical implementation.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to the lead contact, Ruijiang Li (rli2@stanford.edu).

Materials availability

This study did not generate new, unique reagents.

Data and code availability

Publicly available histopathology, multi-omics, and clinical datasets were obtained from multiple sources. TCGA histology slides and clinical annotations were downloaded from the National Cancer Institute Genomic Data Commons Portal: <https://portal.gdc.cancer.gov> and cBioPortal: <https://www.cbioportal.org>. WSIs from the NLST cohort were acquired via The Cancer Imaging Archive (TCIA: <https://wiki.cancerimagingarchive.net>). PLCO datasets were accessed through the Cancer Data Access System (CDAS: <https://cdas.cancer.gov/learn/plco>) following appropriate data request approvals. The CODEX data and matched H&E images generated in this study are deposited in Zenodo: <https://zenodo.org/records/20263843>. The CANVAS pipeline and analysis scripts are publicly available at GitHub: <https://github.com/lilab-stanford/CANVAS>.

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AUTHOR CONTRIBUTIONS

R.L. and Yuchen Li conceptualized and designed the project. Yuchen Li and Z.L. performed data analysis and interpretation and generated the figures. R.Q. provided biological expertise and contributed to data interpretation. Y.J. provided model-related technical expertise. X.Z. provided pathology expertise. J.X., X.W., S.Y., F.E., Y.C., X.L., and S.C. contributed statistical and technical support. Yuanyuan Li performed PCF-related experimental work. J.M., C.B., T.K., F.M.O., S.W., S.H.L., J.J.N., R.W., J.N., and M.D. provided clinical expertise and administrative support. R.L., Yuchen Li, Z.L., and R.Q. wrote the manuscript with input from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
CD8-BX026-Atto 550	Akoya Biosciences	Cat# 4250012; RRID: AB_2915960
Pan-Cytokeratin-BX019-Alexa Fluor™ 488	Akoya Biosciences	Cat# 4150020; RRID: AB_2909509
CD3e-BX045-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550119; RRID: AB_2936080
CD163-BX023-Atto 550	Cell Signaling Technology	Cat# 25121; RRID: AB_3675763
CD20-BX007-Alexa Fluor™ 488	Akoya Biosciences	Cat# 4150018; RRID: AB_2895056
CD4-BX003-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550112; RRID: AB_3094499
FAP-BX052-Atto 550	Abcam	Cat# ab271976; RRID: AB_3676372
CD138-BX016-Alexa Fluor™ 488	Abcam	Cat# ab226108; RRID: AB_3741468
CD11c-BX024-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550114; RRID: AB_3083459
CD66b-BX020-Atto 550	R&D Systems	Cat# MAB4246; RRID: AB_3658275
αSMA-BX013-Alexa Fluor™ 488	Abcam	Cat# ab5694; RRID: AB_2223021
CD68-BX015-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550113; RRID: AB_2935894
Ki67-BX047-Atto 550	Akoya Biosciences	Cat# 4250019; RRID: AB_2895046
CD31-BX001-Alexa Fluor™ 488	Akoya Biosciences	Cat# 4150017; RRID: AB_3678451
Collagen IV-BX042-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550122; RRID: AB_2927676
Granzyme B-BX041-Atto 550	Akoya Biosciences	Cat# 4250055; RRID: AB_3472025
MMP9-BX028-Alexa Fluor™ 488	BioLegend	Cat# 819701; RRID: AB_2564833
PD-1-BX046-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550038; RRID: AB_3096407
CD44-BX005-Atto 550	Akoya Biosciences	Cat# 4450041; RRID: AB_2936081
PD-L1-BX067-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550128; RRID: AB_3676534
E-cadherin-BX014-Atto 550	Akoya Biosciences	Cat# 4250021; RRID: AB_2895057
LAG3-BX055-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550058; RRID: AB_3096409
Mac2/Galectin-3-BX035-Atto 550	Akoya Biosciences	Cat# 4450034; RRID: AB_3477119
FOXP3-BX031-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550071; RRID: AB_2927679
CD14-BX037-Atto 550	Akoya Biosciences	Cat# 4450047; RRID: AB_3083457
EpCAM-BX091-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550088; RRID: AB_2935888
CD21-BX032-Atto 550	Akoya Biosciences	Cat# 4450027; RRID: AB_3474900
CD45-BX021-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550121; RRID: AB_3674468
MPO-BX098-Atto 550	Akoya Biosciences	Cat# 4250083; RRID: AB_2927678
TCF-1-BX061-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550068; RRID: AB_3717468
GATA3-BX049-Atto 550	Akoya Biosciences	Cat# 4250085; RRID: AB_2927677
ICOS-BX054-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550117; RRID: AB_3096408
HLA-A-BX029-Atto 550	Akoya Biosciences	Cat# 4250100; RRID: AB_3094501
Bcl-2-BX085-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550089; RRID: AB_2936078
HLA-E-BX034-Atto 550	Akoya Biosciences	Cat# 4250065; RRID: N/A
CD45RO-BX017-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550127; RRID: AB_3662766
VISTA-BX040-Atto 550	Akoya Biosciences	Cat# 4250063; RRID: N/A
HIF1A-BX062-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550069; RRID: AB_3082972
CD39-BX099-Atto 550	Akoya Biosciences	Cat# 4250076; RRID: AB_3096410
CD40-BX010-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550064; RRID: AB_3676527
HLA-DR-BX033-Alexa Fluor™ 647	Akoya Biosciences	Cat# 4550118; RRID: AB_3080864

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
Formalin-fixed paraffin-embedded (FFPE) human lung tumor samples	This Manuscript	N/A
Chemicals, peptides, and recombinant proteins		
CODEX barcode BX-023 (used for CD163 custom conjugation)	Akoya Biosciences	Cat# 5250003
CODEX barcode BX-052 (used for FAP custom conjugation)	Akoya Biosciences	Cat# 5250012
CODEX barcode BX-016 (used for CD138 custom conjugation)	Akoya Biosciences	Cat# 5150001
CODEX barcode BX-020 (used for CD66b custom conjugation)	Akoya Biosciences	Cat# 5250002
CODEX barcode BX-013 (used for α SMA custom conjugation)	Akoya Biosciences	Cat# 5450017
CODEX barcode BX-028 (used for MMP9 custom conjugation)	Akoya Biosciences	Cat# 5150005
Deposited data		
CODEX and H&E data	This paper	Zenodo: https://zenodo.org/records/20263843
Software and algorithms		
R 4.4.0	R Core Team	https://www.r-project.org
Python 3.10.14	Python Software Foundation	https://www.python.org/downloads/release/python-31014/
CODEX Instrument Manager	Akoya Biosciences	https://help.codex.bio/codex/cim/
CODEX Processor	Akoya Biosciences	https://help.codex.bio/codex/processor/
ImageJ v2.3.0	ImageJ	https://imagej.net/ij/
QuPath v0.4.3	QuPath	https://qupath.github.io/
Seurat 5.1.0	Satija Lab	https://satijalab.org/seurat
Harmony 1.2.1	CRAN	https://cran.r-project.org/web/packages/harmony/index.html
mclust 6.1.1	CRAN	https://cran.r-project.org/package=mclust
ConsensusClusterPlus 1.68.0	Bioconductor	https://bioconductor.org/packages/release/bioc/html/ConsensusClusterPlus.html
GSVA 1.52.3	Bioconductor	https://www.bioconductor.org/packages/release/bioc/html/GSVA.html
survival 3.6-4	CRAN	https://cran.r-project.org/package=survival
survminer 0.5.0	CRAN	https://cran.r-project.org/package=survminer
randomForestSRC 3.3.2	CRAN	https://cran.r-project.org/package=randomForestSRC
CANVAS pipeline and analysis scripts	This paper	GitHub: https://github.com/lilab-stanford/CANVAS
Other		
PhenoCycler Antibody Conjugation Kit	Akoya Biosciences	Cat# 7000009
PhenoCycler Assay Reagent	Akoya Biosciences	Cat# 7000002
10X Buffer Kit for PhenoCycler-Fusion	Akoya Biosciences	Cat# 7000019
CODEX Storage Buffer	Akoya Biosciences	Cat# 232107

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human specimens and study cohorts

This study integrated seven study cohorts to develop, validate, and clinically evaluate the CANVAS framework. A total of 457 lung tumor specimens were profiled by CODEX using a 41-plex antibody panel across three cohorts: tissue microarray cohort 1 (TMA1, $n = 337$; C1), tissue microarray cohort 2 (TMA2, $n = 110$; C2), and a whole-slide imaging cohort (WSI, $n = 10$; C3) (Table S2). TMA1 samples were profiled by CODEX alone, whereas TMA2 and WSI tumors underwent paired CODEX profiling and matched H&E staining. Collectively, these cohorts were used to define cellular neighborhoods from CODEX data. The paired WSI cohort was used to train and validate the H&E-based habitat inference model, which was further evaluated in the independent paired TMA2 cohort. For downstream clinical evaluation using H&E alone, we analyzed three lung cancer cohorts, TCGA ($n = 945$), PLCO ($n = 382$), and NLST ($n = 366$), comprising 1,693 patients (C4) in total (Figure 4C), to assess associations between predicted tumor habitats, patient prognosis, and molecular subtypes. We further extended the analysis to a pan-cancer TCGA cohort spanning eight cancer types ($n = 3,724$; C5) to evaluate cross-cancer generalizability. Finally, two independent immunotherapy-treated cohorts were used for outcome modeling: an internal Stanford ICB cohort ($n = 149$; C6) and a lung cancer cohort from the Cancer Moonshot Biobank ($n = 40$, including 34 PFS-evaluable patients; C7). Stanford specimens were analyzed in accordance with applicable data-use requirements, whereas TMA2 was obtained from TissueArray. Publicly available or controlled-access cohorts, including TCGA, PLCO, NLST, and CMB, were accessed and analyzed in accordance with their respective data-use policies. In summary, CANVAS comprises three integral components: (1) discovery of biologically grounded CNs from CODEX data; (2) inference of CODEX-derived habitats from H&E slides; and (3) derivation of spatial habitat features for prognostic modeling and biological interpretation. A schematic overview of the study design, patient cohorts, and analytical workflow is provided in Figure 1A.

METHOD DETAILS

Sample preparation and tissue sectioning

Formalin-fixed paraffin-embedded (FFPE) tumor specimens from patients with NSCLC were used for CODEX imaging. For the Stanford TMA1 cohort, tumor regions were identified by pathology review. Representative tumor regions were sampled from archival donor blocks using tissue punches and arrayed into recipient TMA blocks. In addition, surgical FFPE tumor blocks from patients with NSCLC were sectioned as whole-slide specimens to construct the Stanford WSI cohort. For CODEX staining, 5- μ m FFPE sections from TMAs and whole-slide specimens were mounted onto Fisherbrand Superfrost Plus microscope slides (Fisher Scientific) according to the PhenoCycler-Fusion User Guide v2.1.0 and processed for downstream staining.

CODEX staining, imaging, and cell segmentation

Spatial proteomic profiling was performed at the Cell Sciences Imaging Facility (CSIF) at Stanford University using the PhenoCycler-Fusion (PCF) 2.0 platform, formerly known as CODEX. The antibody panel included pre-conjugated CODEX antibodies obtained from Akoya Biosciences and custom-conjugated antibodies generated using the PhenoCycler Antibody Conjugation Kit (Akoya Biosciences) according to the manufacturer's protocol. Antibody staining specificity and conjugation performance were routinely validated by immunofluorescence and PhenoCycler-Fusion test runs. Staining conditions were optimized individually for each antibody. Antibodies used in this study are listed in Table S1. FFPE sections from TMAs and whole-slide specimens were stained using the PhenoCycler Staining Kit. Briefly, after overnight baking at 60°C, slides were deparaffinized using a standard histological procedure. Antigen retrieval was conducted in Tris-EDTA buffer (pH 9.0) in a pressure cooker at 120°C for 20 min under high pressure. After cooling to room temperature, slides were incubated in a photobleaching buffer for 45 min. After extensive washing in 1 $\times$ PBS, slides were hydrated and incubated with the conjugated antibody cocktail overnight at 4°C in a humidity chamber, followed by post-staining fixation and washing steps as instructed in the manufacturer's user manual. Tissue sections were stored in CODEX Storage Buffer at 4°C until imaging. Reporter and buffer reagents for PhenoCycler-Fusion imaging were prepared using PhenoCycler Assay Reagent and the 10X Buffer Kit for PhenoCycler-Fusion (Akoya Biosciences) according to the manufacturer's instructions. High-plex immunofluorescence imaging was then performed on the PhenoCycler-Fusion 2.0 platform. Images were acquired in DAPI, AF488, Atto550, and AF647 channels and automatically processed into multiplexed image files for downstream analysis.

Multiplexed fluorescence images were acquired using the CODEX Instrument Manager at a resolution of 0.507 μ m per pixel. Raw image stacks were processed through the CODEX Processor pipeline for cross-cycle image alignment, tile stitching, and autofluorescence subtraction. Processed CODEX images were reviewed in ImageJ (v2.3.0) and QuPath (v0.4.3), and high-quality images were selected for downstream analysis. Each TMA core was treated as a single region of interest (ROI). Processed image stacks subsequently underwent standardized preprocessing and quality control, including tissue detection, ROI curation, and single-cell segmentation on the DAPI channel.⁸⁴ Nuclear expansion was performed to approximate whole-cell boundaries. Marker expression was quantified as the mean fluorescence intensity within each segmented cell mask. Shape and intensity features were computed for segmented cell compartments, and probability maps were retained to support downstream quality control. In total, over 18 million cells were analyzed across all samples, including over 1.4 million cells from the TMA cohorts and over 16.6 million cells from the WSI cohort.

H&E staining and imaging

Following PhenoCycler imaging, flow cells were removed according to the manufacturer's instructions. Slides were washed extensively with $1 \times$ PhenoCycler buffer to remove residual imaging reagents and then processed for post-CODEX H&E staining using a standard histopathology workflow. Briefly, sections were rinsed, stained with hematoxylin, counterstained with eosin, dehydrated, cleared, and coverslipped. H&E-stained sections were digitized using a Leica Aperio AT2 whole-slide scanner at $40\times$ magnification.

Cell type and cell state annotation using CODEX data

Single-cell spatial proteomic data were generated from pixel-level quantification of marker intensities and processed using the Seurat⁸⁵ package for downstream analysis. To reduce signal variation across markers, data normalization was performed using *NormalizeData*, followed by feature scaling with *ScaleData*. Dimensionality reduction was performed by principal component analysis using *RunPCA*, and sample-level batch effects were corrected with the R package Harmony using *RunHarmony*. UMAP visualization was generated from the Harmony-corrected embedding, and graph-based clustering was performed using *FindNeighbors* and *FindClusters*. Cluster-enriched markers identified using *FindAllMarkers* were used to assign major lineage identities. Cells with potential segmentation artifacts, ambiguous marker profiles, or clusters lacking clear lineage-defining marker expression were annotated as unknown and were handled separately from annotated cell populations in subsequent analyses. CD45⁺CD3e⁺GZMB⁺ immune cells were annotated as NK-like cells based on the available marker panel and referred to as NK cells in downstream analyses. Detailed functional cell states were annotated by integrating high-resolution clustering with targeted Gaussian mixture modeling⁸⁶ for selected marker-defined populations, using curated marker combinations derived from the literature. Given the finite marker space of the CODEX panel, a subset of detailed annotations was assigned based on marker-informed phenotypic patterns. For whole-slide CODEX datasets, a two-stage annotation strategy was used. A representative subset of cells from each sample was first annotated to generate a reference set, and anchor-based label transfer was then performed for the remaining cells using *FindTransferAnchors* and *TransferData*.

Pairwise cell-cell interaction and co-occurrence

To quantify pairwise cell-cell interactions, we applied a radius-based, permutation-derived spatial interaction framework in which all neighboring cells within $40\ \mu\text{m}$ of each cell centroid were defined as the local microenvironment.⁸⁷ Pairwise interactions between annotated cell phenotypes were assessed independently within each image by comparing the observed frequency of neighboring cell-type pairs with a permutation-based null distribution. To generate the null model, cell phenotypes were randomized 1,000 times while preserving the underlying spatial coordinates. Cell-type pairs enriched relative to the permuted background were interpreted as spatial attraction, whereas depleted pairs were interpreted as spatial avoidance. By evaluating interactions within fixed local neighborhoods, this analysis captures short-range spatial organization and immediate multicellular topology in the TME.

Cell-cell co-occurrence was quantified at the tissue-core level to assess whether pairs of cell phenotypes were jointly enriched across samples, independent of their physical proximity within individual tissues. This analysis reflects coordinated expansion or depletion of lineages at the tissue scale, revealing global immune or stromal equilibria not detectable from local interactions alone.

Identification of spatial cellular neighborhoods

To characterize multicellular spatial niches within the tumor microenvironment, we applied spatial Latent Dirichlet Allocation (spatial-LDA),^{87,88} a topic modeling framework adapted for spatially resolved cell phenotype data. For each index cell, a local neighborhood was defined as all cells within a $40\ \mu\text{m}$ radius, a scale that consistently captured ~ 25 neighboring cells across cohorts and aligns with prior studies delineating spatial domains in tissue microenvironments.^{89,90}

Spatial-LDA was applied to model local cell composition as mixtures of latent motifs enriched for specific cell types, with spatial neighborhoods defined to retain local tissue context. The resulting motif proportion profiles were subjected to K-means clustering to define CNs, which capture spatially coherent and biologically distinct multicellular ecosystems within the tissue microenvironment. To improve cross-cohort independence and cross-format transferability, CNs were derived by co-clustering each TMA cohort separately with the WSI cohort. To determine the optimal number of CNs, we evaluated candidate solutions ($k = 5\text{--}20$) using complementary validity and stability metrics, including the Silhouette coefficient, Davies-Bouldin index, and Adjusted Rand index calculated between adjacent K partitions.

Graph-based triplet motif analysis

Spatial cell-cell interaction graphs were constructed for each TMA core by connecting CODEX-derived single-cell centroids within a fixed local distance threshold. Each cell was assigned to one of predefined cell types based on single-cell annotations. Local graphs from individual cores were aggregated into a global cell-type interaction network, where nodes represented cell types and edge weights corresponded to pairwise interaction frequencies. To quantify the topological prominence of each cell type, we employed a graph neural network (GNN) approach, implemented using a graph convolutional network (GCN).⁹¹ The model used graph topology and interaction weights through a GCN-based architecture to derive node-level topological scores. It was optimized to minimize the mean squared error between predicted and observed node degrees. The resulting node GCN scores captured each cell type's topological prominence within the global interaction topology.

All possible three-cell-type combinations were exhaustively enumerated, yielding 455 candidate triplet motifs. For each triplet, global network-informed scoring metrics were derived from edge-weight and GCN-based measures of the constituent cell types. Rankings from these metrics were integrated using robust rank aggregation (RRA), and 24 high-confidence triplet motifs with $FDR < 0.01$ were retained for downstream analysis. At the core level, each retained motif was summarized based on the local spatial organization of its constituent cell types, and motif-level scores were used for subsequent association analyses.

Proximity density and barrier score quantification

Proximity density (PD) quantifies the normalized local interaction density between two specified cell types within a given CN domain, defined as the number of identified proximity interactions divided by the total abundance of those two cell types within the same CN.⁸⁷ This metric reflects localized cell-type juxtaposition and captures the extent of cell-type co-engagement within spatially constrained tissue niches.

To quantify stromal-mediated immune exclusion, we adapted the concept of a barrier score from the topological tumor graph framework.⁹² CD8⁺ T cells, proliferating tumor cells, and CAFs were represented in a spatial graph. For each cytotoxic T cell not directly adjacent to a tumor cell, shortest immune–tumor paths were identified, and each path was classified as barrier-positive if at least one CAF was present along the intervening path. The sample-level barrier score was defined as the fraction of barrier-positive shortest immune–tumor paths across eligible cytotoxic T cells, with higher scores indicating stronger stromal insulation of tumor cells from cytotoxic T cell access.

H&E image preprocessing, tumor detection, and tissue and cell segmentation

All H&E whole-slide images were analyzed at a standardized resolution equivalent to 40× magnification, corresponding to 0.25 micrometers per pixel. Slides acquired at lower resolutions were digitally rescaled to this reference magnification to ensure uniformity across cohorts. Regions affected by technical artifacts, including pen markings, tissue folds, and image blur, were systematically excluded using automated color-based filtering algorithms.⁹³ Tissue-containing areas were segmented using the CLAM pipeline,⁹⁴ followed by extraction of non-overlapping patches of 224 by 224 pixels. H&E images were color-normalized to reduce inter-slide staining variability. Tumor area detection was performed using a vision transformer model initially pretrained on a pan-cancer histopathology dataset and subsequently fine-tuned for tumor detection.^{95,96} Tumor-region detection generated patch-level tumor probability scores, which were spatially aggregated to identify tumor-enriched regions for downstream analysis. Cellular detection and segmentation in all datasets were performed using CellViT,⁹⁷ a transformer-based model optimized for precise nuclear segmentation in H&E-stained tissues. The resulting cell coordinate matrices provided the basis for multimodality co-registration and habitat inference.

To define tumor bulk and leading-edge regions on H&E-stained whole-slide images, patch-level tumor probability maps were spatially aggregated to capture regional tumor context. These neighborhood-level features were subjected to unsupervised spatial clustering, and the resulting clusters were annotated as tumor bulk, leading-edge, or other tissue compartments based on tumor probability patterns and spatial organization. The annotated regions were then used for downstream habitat profiling and prognostic modeling.

Multi-modality image co-registration

To facilitate deep learning between CODEX and H&E images, we implemented a pipeline for high-precision single-cell co-registration. Specifically, the DAPI channel from CODEX was aligned with the hematoxylin channel of the corresponding H&E section using PALOM (PyPI: <https://pypi.org/project/palom/>). In this framework, the source coordinates were defined as the centroid positions of segmented single cells from CODEX images. Affine transformation parameters, including translation, rotation, scaling, and shear, were used to project CODEX-derived cell coordinates into H&E space. This spatial alignment established the correspondence between different image modalities, enabling direct transfer of CODEX-defined CN annotations onto H&E histopathology.

Formally, given a source coordinate $\mathbf{x}_{src} = [x, y, 1]^T$, the corresponding destination coordinate in H&E space $\mathbf{x}_{dst} = [x_{dst}, y_{dst}]^T$ was computed using an affine matrix $\mathbf{A} \in \mathbb{R}^{2 \times 3}$ as:

$$\mathbf{x}_{dst} = \mathbf{A} \cdot \mathbf{x}_{src},$$

where:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \end{bmatrix},$$

and t_x, t_y represent translation, while a_{ij} encode rotation, scaling, and shear components.

Annotations that mapped outside the spatial boundaries of the H&E images were excluded from further analysis. Registration quality was manually reviewed to confirm alignment of corresponding tissue structures across modalities. H&E-derived single-cell segmentations were assigned CN labels by reference to the nearest registered CODEX cell centroid, using a centroid-to-centroid

distance threshold of 5 μm . Only cells successfully mapped to one of ten spatial habitats were retained for downstream modeling. This image co-registration pipeline established high-confidence ground-truth habitat labels to learn spatially encoded morphological patterns from H&E.

Habitat label transfer based on CN assignment

CNs were derived by co-clustering the WSI cohort with TMA1 for discovery and with TMA2 for independent validation. CN labels were transferred to co-registered H&E images as habitat annotations for downstream model training and evaluation. Patch-level training labels were derived from CODEX-based single-cell annotations that were spatially mapped onto the corresponding H&E tissue sections. Patches with evident technical artifacts, including tissue folds, blurred regions, staining artifacts, or image misregistration, were manually excluded during quality control. For each image patch used for model training, the number of cells assigned to each spatial habitat class was quantified. Patches containing five or fewer annotated cells were assigned to a background class to represent non-informative tissue regions. Among habitat-informative patches, only those with more than fifteen annotated cells and a dominant habitat class accounting for at least 60% of the local cellular composition were retained as high-confidence habitat-labeled patches. Each eligible patch was labeled according to its most abundant habitat class. This filtering strategy emphasized the selection of high-confidence and compositionally consistent regions to enhance the reliability of model supervision. A multimodal dataset of spatially co-registered CODEX/H&E image patches (224×224 pixels) was assembled for model development. Following quality control, WSI patches were assigned to one of ten CODEX-derived CN labels and split into training ($>180\text{K}$ patches) and validation ($>82\text{K}$ patches) sets.

CANVAS model architecture and training strategy

The CANVAS model was developed to classify H&E image patches (224×224 pixels) into one of ten habitat types or a background class, enabling spatial habitat inference across whole-slide images. The model's backbone is based on MUSK,⁷⁸ a foundation model pre-trained using unified masked modeling on over 50 million pathology images from 11,577 patients and one billion pathology-related text tokens. MUSK enables robust extraction of high-dimensional visual embeddings that capture complex morphological features across diverse tissue types and staining variations. For model input, these 224×224 -pixel patches were resized to 384×384 pixels before being passed to the MUSK encoder. Let $\mathbf{h} \in \mathbb{R}^d$ denote the visual feature embedding extracted by the MUSK encoder. The classification head consists of three sequential transformations:

$$\mathbf{z}_1 = \text{Dropout}(\text{ReLU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1)), \mathbf{z}_1 \in \mathbb{R}^{256}$$

$$\mathbf{z}_2 = \text{Dropout}(\text{ReLU}(\mathbf{W}_2 \mathbf{z}_1 + \mathbf{b}_2)), \mathbf{z}_2 \in \mathbb{R}^{128}$$

$$\mathbf{o} = \mathbf{W}_3 \mathbf{z}_2 + \mathbf{b}_3, \mathbf{o} \in \mathbb{R}^{11},$$

where $\mathbf{W}_i$ and $\mathbf{b}_i$ are learnable weights and biases of each layer, and $\mathbf{o}$ represents the unnormalized logits corresponding to the ten habitat classes and a background class.

To address class imbalance, we combined weighted random sampling with focal loss.⁹⁸ For sampling, class frequencies f_{c_i} were computed across the training set, and each sample i (label c_i) received a weight $\frac{1}{f_{c_i}}$. Samples were drawn proportionally to these weights to ensure balanced exposure across classes. In parallel, focal loss was used to reduce the contribution of easy examples and emphasize underrepresented or misclassified instances. Given the predicted probability p_t for the true class, the focal loss is defined as:

$$\mathcal{L}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t).$$

The classification objective was optimized using the focal loss function, where α denotes the focal-loss weighting factor, and γ modulated the contribution of easy-to-classify samples to focus learning on harder examples. To enhance model robustness against staining and morphological variability, real-time data augmentation was implemented during training. Augmentation procedures included random horizontal and vertical flipping, rotation, and perturbation of image brightness, contrast, saturation, and hue. All augmented images were subsequently converted to tensors and normalized using channel-wise means and standard deviations used by the MUSK preprocessing pipeline. This preprocessing pipeline improved model generalization and mitigated overfitting, particularly for underrepresented habitat classes.

Model evaluation and performance benchmarking

The CANVAS model was trained and internally validated using the Stanford WSI cohort, which was randomly partitioned into 80% training and 20% validation sets at the sample level. TMA2 served as an independent external test set to evaluate model classification performance across histologic platforms. The model was trained using the Adam optimizer. The initial learning rate was set to 1×10^{-4} , with exponential decay applied at a factor of 0.95. A batch size of 64 and dropout rate of 0.5 were used throughout training. The model was trained for 60 epochs using a two-stage fine-tuning approach. For the first 10 epochs, only the classification head was trained while the entire backbone remained frozen. Subsequently, for the remaining 50 epochs, the final two layers of the encoder were

unfrozen and fine-tuned along with the classification head. Evaluation was conducted using patch-level classification accuracy and habitat-type-specific precision and recall, measured on a held-out validation set with no patient overlap. Model performance was further validated across multiple cohorts via whole-slide inference and spatial prediction of habitats.

To evaluate the predictive performance of CANVAS-based habitat classification models, we computed three standard metrics: overall accuracy, macro-averaged F1 score, and Cohen's kappa coefficient.

Accuracy was calculated as the proportion of correctly predicted habitat labels among all predictions:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}.$$

F1 score was computed as the harmonic mean of precision and recall for each habitat class, and macro-averaged across all classes:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Macro-F1} = \frac{1}{K} \sum_{i=1}^K F1_i.$$

Cohen's kappa coefficient quantified inter-rater agreement between predicted and true labels beyond chance, defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o denotes the observed agreement and p_e represents the expected agreement by random chance.

We benchmarked the performance of CANVAS against three baselines: two pathology foundation models (UNI⁹⁹ and Virchow¹⁰⁰) and ResNet-50. All models were trained and evaluated under identical conditions with standardized preprocessing, patient-level splits, and consistent label definitions. Predictive performance was quantified by Macro-F1 and Cohen's kappa. Robustness of performance differences was assessed by 1,000 bootstrap resampling of the validation and testing cohorts.

Evaluation of resolution and stride overlap in habitat prediction

The impact of image resolution on model performance was assessed by systematically downsampling WSIs and TMA cores from the validation and test sets. Slides originally scanned at 40× magnification (0.25 μm per pixel) were downsampled to 20× (0.50 μm per pixel) and 10× (1.00 μm per pixel). Habitat predictions were then generated using the trained CANVAS model, and performance was quantified using standard patch-level classification metrics.

To evaluate the effect of tiling strategies on CANVAS habitat predictions, we varied the stride to modulate overlap between adjacent tiles, thereby testing whether increased sampling density around tumor boundaries or immune-enriched regions influenced model outputs. The patch size was fixed at 224 × 224 pixels, and three stride conditions were tested: 224 pixels (non-overlapping tiles), 112 pixels (50% overlap), and 56 pixels (75% overlap). For each stride setting, patch datasets were generated separately for the test set, and habitat predictions were performed. Model performance was quantified using accuracy, and robustness was further assessed by nonparametric bootstrap resampling with 1,000 iterations across stride pairs (224 vs. 112 pixels; 224 vs. 56 pixels).

Prognostic modeling of habitats on H&E-stained images

To evaluate the prognostic significance of CANVAS-derived spatial habitats, we quantified the habitat profiles of 10 habitats within the tumor bulk and leading-edge compartments of each sample across the TCGA, PLCO, and NLST cohorts. Samples with insufficient patch coverage in both tumor-bulk and leading-edge regions were excluded to ensure adequate spatial representation. Within each histologic subtype and cohort, habitat–outcome associations were evaluated separately using available clinical endpoints, including overall survival (OS), disease-specific survival (DSS), progression-free interval (PFI), and disease-free interval (DFI). For each habitat–compartment combination, univariate Cox proportional hazards models were fitted, and hazard ratios (HRs), 95% confidence intervals, and Wald *P* values were reported.

Spatial ecotypes identified using H&E-derived tumor habitats

To identify recurrent spatial phenotypes with prognostic relevance, we performed unsupervised consensus clustering of TCGA-LUAD samples based on CANVAS-derived habitat profiles in tumor bulk and leading-edge regions. A feature matrix comprising the distribution of 10 habitats across the two spatial compartments was z-score normalized and subjected to consensus clustering using the ConsensusClusterPlus R package¹⁰¹ with partitioning around medoids (PAM) and Canberra distance. The resulting consensus-defined subtypes were subsequently linked to clinical variables and survival outcomes. Prognostic value was further assessed using multivariable Cox proportional hazards models adjusted for major clinical covariates, including age, sex, smoking status, and tumor stage. To further contextualize the identified spatial ecotypes, we compared them with transcriptomic TME subtypes assigned to matched samples.³⁸

Orthogonal validation of habitat prediction

We assembled orthogonal validation datasets comprising same-section CODEX–H&E specimens and paired Xenium–H&E specimens (Table S2). For Xenium data, preprocessing, clustering, and cell-level annotation were performed separately for each sample using a common Seurat workflow that included quality control, *SCTransform* normalization, dimensionality reduction, and sample-level clustering. CN labels were inferred at single-cell resolution using a cosine-similarity-based CN inference framework, in which cell-level neighborhood profiles from spatial omics data were mapped to the most similar CODEX-derived reference CN composition profiles.

Gene set enrichment and immunogenomic analysis

To characterize the tumor microenvironmental correlates of CANVAS-defined spatial habitats, we computed patient-level habitat profiles for each TCGA-NSCLC sample by aggregating patch-level predictions of H01–H10. Samples with limited informative patch coverage were excluded. These spatial features were integrated with three orthogonal TME inference frameworks: (1) ESTIMATE¹⁰²-derived stromal, immune, and tumor purity scores; (2) CIBERSORTx (LM22)^{103,104} immune cell scores; and (3) xCell¹⁰⁵ enrichment profiles spanning 64 cell-type signatures. All data were harmonized using TCGA sample barcodes to enable multi-omic correlation analysis. The GSVA package¹⁰⁶ was used to compute pathway-level enrichment scores from normalized transcriptomic data, leveraging curated gene sets from MSigDB, KEGG, Reactome, and other collections.³⁸ Separately, ecotype-associated pathway enrichment across CANVAS-inferred spatial ecotype groups was assessed using the limma¹⁰⁷ R package with empirical Bayes moderation and linear modeling to rank genes for preranked GSEA.

Genomic and immunologic correlates, including tumor mutational burden, aneuploidy, microsatellite instability, and SNV-derived neoantigen load, were obtained from public databases. Transcript-based signatures, including interferon response, tertiary lymphoid structure, EMT-TGF β signaling, and angiogenesis signatures, were compiled from the literature and calculated using ssGSEA as implemented in the GSVA package. Associations between spatial ecotypes and molecular features were assessed using covariate-adjusted analysis of variance, with age and sex included as clinical covariates, implemented using the Anova function from the car¹⁰⁸ R package.

Somatic mutation and copy number alteration association analysis

To investigate genomic correlates of CANVAS-defined spatial ecotypes, gene-level copy number alteration and somatic mutation data from the TCGA-LUAD cohort were analyzed. Gene-level copy number calls were obtained from TCGA/cBioPortal processed data and encoded as deletion, neutral, or amplification events. Associations between gene-level copy number status and spatial ecotype membership were assessed using regression-based models. Effect sizes were summarized as odds ratios with corresponding 95% confidence intervals. To facilitate biological interpretation, candidate genes were further annotated according to established oncogenic and immunoregulatory pathways.¹⁰⁹

Somatic mutation profiles were obtained from cBioPortal-derived MC3 consensus calls and binarized at the gene level. For each retained gene, mutation frequencies were calculated within each spatial ecotype, and associations between mutation status and ecotype membership were assessed using Fisher's exact tests. Recurrent ecotype-associated mutation patterns were visualized by heatmap after z-score normalization of mutation frequencies. Candidate genes were curated from established TCGA PanCancer Atlas driver gene resources.¹¹⁰

Habitat-level spatial feature engineering and extraction

To biologically interpret CANVAS-defined spatial habitats, we computed a total of 262 quantitative features from H&E-stained NSCLC tissue sections. All features were derived from patch-level annotations and subsequently aggregated to the image level to enable cohort-wide comparative analyses. **(1) Composition features (n = 10):** patch-level CANVAS annotations were aggregated per image to quantify the frequency of each of the ten spatial habitats. **(2) Diversity features (n = 6):** to capture ecological complexity within each image, we computed six diversity metrics using habitat label counts as species abundances, including richness, Shannon index, Simpson index, inverse Simpson index, Fisher's alpha, and Pielou's evenness. **(3) Spatial dispersion features (n = 90):** to characterize intra-habitat spatial organization, we converted habitat-specific coordinates into planar point patterns and derived a set of spatial summary features for each habitat, including summaries from Ripley's K and L functions, pair correlation, G, F, and J functions, together with the Clark–Evans index, quadrat-based dispersion statistics, and kernel density summaries. **(4) Interaction features (n = 100):** to evaluate inter-habitat spatial coupling, pairwise interaction scores were computed for each tissue image using a fixed-radius neighborhood framework. For each habitat pair, interaction scores were derived using a permutation-based scoring approach (1,000 iterations), comparing observed spatial adjacency frequencies with a randomized null distribution. This yielded 100 ordered habitat-pair interaction features corresponding to the full 10 $\times$ 10 interaction matrix. **(5) Distance features (n = 55):** pairwise nearest-neighbor Euclidean distances between habitat pairs were calculated for each tissue image and summarized as image-level spatial distance features. **(6) Transition feature (n = 1):** Spatial transition entropy (STE) was computed to quantify the habitat transition complexity within each image. For each tissue image, habitat transitions were defined based on patch-level *k*-nearest-neighbor relationships (*k* = 6), and a habitat transition matrix was constructed from all observed transitions between neighboring patches. STE was then calculated as the Shannon entropy of the resulting global transition probability distribution.

Feature selection and model development

To construct a robust model for predicting immunotherapy outcomes, we implemented a multi-stage computational pipeline encompassing feature curation, redundancy reduction, feature prioritization, and survival modeling. As an unsupervised preprocessing step, multicollinearity was first reduced by calculating the Spearman correlation matrix across all predictors. Features with absolute pairwise correlation > 0.95 were grouped by Louvain community detection using the *igraph* package,¹¹¹ and one representative feature from each cluster was retained to generate a reduced, non-collinear feature set.

Robust candidate predictors were then prioritized using two complementary resampling-based approaches. First, we performed 100 repeated resampling iterations of LASSO-penalized Cox regression using the *glmnet* package.¹¹² In each iteration, patients were randomly divided into training and validation subsets, and the training subset was used for model fitting with internal 5-fold cross-validation to determine the optimal penalty parameter based on the concordance index. Features ranked in the top 10% by selection frequency were retained. Second, we applied permutation-based random survival forest analysis using the *randomForestSRC* package. Across 1,000 bootstrap iterations, feature importance was quantified by permuting individual predictors and measuring the resulting change in prediction error for progression-free survival. Features showing consistently high permutation importance were prioritized as robust predictors.

Thirteen features jointly prioritized by the LASSO and random survival forest procedures were subsequently used to train a multi-variable Cox proportional hazards model. Model coefficients were estimated using the available cohort and model performance was evaluated using the C-index and time-dependent AUCs at 6, 12, and 24 months. For survival visualization, patients were stratified into risk groups based on the median risk score within the corresponding cohort. Prognostic separation was summarized using hazard ratios, log-rank P values, and 95% confidence intervals. Final model coefficients were visualized in a forest plot with annotation categories.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were conducted using R (version 4.4.0) and Python (version 3.10) for data preprocessing, visualization, and hypothesis testing. Two-group comparisons were performed using two-sided t tests or Mann–Whitney U tests, as appropriate. For comparisons involving more than two groups, ANOVA or non-parametric Kruskal–Wallis tests were applied as appropriate. P values were adjusted for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure where applicable. Survival analysis was performed using Kaplan–Meier estimators with log-rank tests, and survival associations were further evaluated using univariate and multivariable Cox proportional hazards models implemented in the *survival* R package. Unless otherwise specified, all tests were two-sided, and $P < 0.05$ was considered statistically significant.